Group Actions on Hyperbolic Spaces

Lecture Notes

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Part 1

Introduction

LECTURE 1

Metric spaces, trees, and free groups

1.1. Motivation and metric spaces

Goal. Study groups G (usually finitely presented) via the geometry of their actions. Finitely presented groups are countable and discrete, so tools from the theory of topological groups are not directly useful. Instead we focus on making the group act on a metric space.

- If a group has a sufficiently nice action on a metric space, the geometry of that space influences the group's properties.
- Negative curvature (hyperbolicity) is a group property.
- Hyperbolic groups have a well-defined boundary at infinity.

dynamics on the boundary	$\longleftrightarrow$	geometry of the space
	$\longleftrightarrow$	algebraic structure of the group

DEFINITION 1.1.1. A *metric space* (X, d) consists of a set X and a function $d\colon X \times X \to \mathbb{R}$ satisfying, for all $x, y, z \in X$:

Positive definiteness: $d(x, y) \geq 0$, with equality if and only if $x = y$.
Symmetry: $d(x, y) = d(y, x)$
Triangle inequality: $d(x, z) \leq d(x, y) + d(y, z)$

The metric topology has as a basis the open balls:

$$B_r(x) := \{y \in X \mid d(x, y) < r\}$$

Also define the closed balls and open and closed neighborhoods of subsets:

$$\bar{B}_r(x) := \{y \in X \mid d(x, y) \leq r\} \qquad N_r(Y) := \{x \in X \mid d(x, Y) < r\} \qquad \bar{N}_r(Y) := \{x \in X \mid d(x, Y) \leq r\}$$

In the latter two we mean $d(x, Y) := \inf_{y \in Y} d(x, y)$.

1.2. Paths, length spaces, and geodesics

A *path* is a continuous map $\gamma\colon I \to X$, where $I \subseteq \mathbb{R}$ is an interval.
The *length* of a path $\gamma\colon [a, b] \to X$ is:

$$|\gamma| := \sup \left\{ \sum_{i=0}^{n-1} d(\gamma(t_i), \gamma(t_{i+1})) \;\middle|\; a = t_0 < t_1 < \cdots < t_n = b \right\}$$

A path is *rectifiable* if its length is finite, and *locally rectifiable* if its restriction to every compact subinterval of its domain has finite length.

LEMMA 1.2.1 (Properties of length). *For a locally rectifiable path $\gamma\colon I \to X$ and $r \leq s \leq t$ in I:*

- $d(\gamma(s), \gamma(t)) \leq \left|\gamma|_{[s,t]}\right|$
- $\left|\gamma|_{[r,s]}\right| + \left|\gamma|_{[s,t]}\right| = \left|\gamma|_{[r,t]}\right|$
- *There is a* unit-speed reparametrization*: For each $[s, t] \subseteq I$ there is a continuous, weakly monotonic function $\phi\colon [s, t] \to \left[0, \left|\gamma|_{[s,t]}\right|\right]$ and a unique path $\tilde{\gamma}\colon \left[0, \left|\gamma|_{[s,t]}\right|\right] \to X$ such that $\tilde{\gamma} \circ \phi = \gamma$ and $\left|\tilde{\gamma}|_{[0,u]}\right| = u$ for all $u \in \left[0, \left|\gamma|_{[s,t]}\right|\right]$.*

DEFINITION 1.2.2. A metric space (X, d) is a *length space* if, for all $x, y \in X$ and every $\varepsilon > 0$, there exists a rectifiable path γ from x to y of length less than $d(x, y) + \varepsilon$.

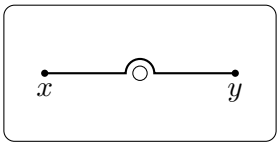
A *geodesic* is a locally rectifiable path $\gamma\colon I \to X$ such that $\forall s, t \in I$:

$$\left|\gamma|_{[s,t]}\right| = d(\gamma(s), \gamma(t))$$

DEFINITION 1.2.3. A metric space (X, d) is a *geodesic space* if every pair $x, y \in X$ can be joined by a geodesic.

EXAMPLE 1.2.4. A space that is not path-connected is not a length space.

EXAMPLE 1.2.5 (Punctured plane). Consider the plane minus a point, $\mathbb{R}^2 \setminus \{\vec{0}\}$, with the restriction of the Euclidean metric. For points x, y whose Euclidean segment passes through the puncture, any path between them in the punctured plane must detour around the puncture, which costs some additional length. We can make the detour as small as desired, but not zero. This space is a length space but not a geodesic space.



1.3. Graphs and trees

A *graph* Γ consists of a vertex set $V\Gamma$ and an edge set $E\Gamma$. Multiple edges and loops are allowed. Regard every edge as having length 1; the induced path-length distance is a metric precisely when Γ is connected, and then it is geodesic.

An *edge path* has the form:

$$v_0 \underset{e_1}{\text{—}} v_1 \underset{e_2}{\text{—}} \cdots \underset{e_n}{\text{—}} v_n$$

A *cycle* is a closed edge path ($v_0 = v_n$) without repeated edges or other repeated vertices.

DEFINITION 1.3.1. A *tree* is a nonempty connected graph without cycles.

LEMMA 1.3.2. *A finite tree that is not a single vertex has at least two* leaves*, that is, vertices of valence* 1*.*

PROOF. Choose a longest embedded edge path $v_0 \underset{e_1}{\text{—}} \cdots \underset{e_n}{\text{—}} v_n$. We claim that v_0 and v_n are leaves. If, say, v_0 had another incident edge f with opposite endpoint w, then either $w \notin \{v_0, \ldots, v_n\}$, giving a longer embedded path, or $w \in \{v_0, \ldots, v_n\}$, giving a cycle. □

COROLLARY 1.3.3. *If Γ is a tree, then every nonconstant closed edge path backtracks at least twice (counted cyclically, so allowing the possibility that one of the backtracks is at the initial/terminal vertex).*

PROOF. The image is a connected finite subtree, hence has at least two leaves. At each leaf the path must backtrack. □

COROLLARY 1.3.4. *If Γ is a tree and $x \neq y$ are vertices, there is a unique embedded edge path from x to y.*

PROOF. Suppose α, β are embedded paths from x to y. Let $\bar{\beta}$ be the edge path β traversed in the opposite direction. The concatenation $\alpha + \bar{\beta}$ is a closed edge path, so it has at least two backtracks. Since α and β have no internal backtracking, these must occur at the endpoints x, y. Removing the common initial and terminal edges and inducting on $d(x, y)$ gives $\alpha = \beta$. □

LEMMA 1.3.5. *For a connected graph Γ, the following are equivalent:*

(i) Γ *is a tree.*
(ii) Γ *has no embedded circle* $\mathbb{S}^1$*.*
(iii) Γ *has no loops and every pair of vertices is joined by a unique embedded edge path.*

(iv) Every pair of points is joined by a unique geodesic.
(v) For each point x, either $\Gamma = \{x\}$, or x is a leaf, or x is a cut point of Γ.

For points x, y of a tree, write $[x, y]$ for the unique geodesic joining them.

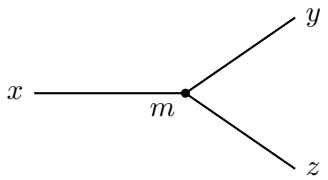
LEMMA 1.3.6. *If Γ is a tree, $x, y \in \Gamma$, and γ is any path from x to y, then the image of γ contains $[x, y]$.*

PROOF. The image of γ contains x and y. If $x = y$ we are done. Otherwise, every interior point z of $[x, y]$ is a cut point of Γ. If z is an interior point of an edge then locally it only has two complementary components, so these identify the two global complementary components. Thus, x and y are separated by z, meaning that every path from x to y passes through z. Since the image of γ is closed and contains the interior points of every edge on $[x, y]$, it also contains the vertices. □

LEMMA 1.3.7 (Tripod lemma). *For any x, y, z in a tree there exists a unique point m such that:*

$$[x, y] \cap [y, z] \cap [x, z] = \{m\}$$

PROOF. Let m be the last common point of $[x, y]$ and $[x, z]$. The two geodesic segments $[y, m]$ and $[m, z]$ concatenate without backtracking, since a backtrack would contradict the choice of m. Thus their concatenation is $[y, z]$, and the triple intersection is exactly $\{m\}$.



□

THEOREM 1.3.8. *Geodesic triangles in a tree are* 0-thin*: if $x_0, x_1, x_2 \in \Gamma$ and γ_i is a geodesic from x_i to x_{i+1} (indices modulo 3), then for all i:*

$$\gamma_i \subseteq \gamma_{i+1} \cup \gamma_{i+2} = \bar{N}_0(\gamma_{i+1} \cup \gamma_{i+2})$$

PROOF. Every triangle is a tripod by the preceding lemma. If m is its central point, then for each i:

$$[x_i, x_{i+1}] = [x_i, m] \cup [m, x_{i+1}] \subseteq [x_i, x_{i+2}] \cup [x_{i+2}, x_{i+1}]$$ □

1.4. The Gromov product and the boundary of a tree

DEFINITION 1.4.1. The *Gromov product* of x and y based at z is:

$$(x \mid y)_z := \frac{1}{2}\left(d(x, z) + d(y, z) - d(x, y)\right)$$

If T is a tree and m is the center of the tripod with vertices x, y, z, then

$$(x \mid y)_z = d(m, z) = d(z, [x, y])$$

Let T be a possibly unbounded tree. A (geodesic) *ray* is an embedded edge path that is infinite in one direction:

$$v_0 \underset{e_1}{\text{—}} v_1 \underset{e_2}{\text{—}} v_2 \text{—} \cdots$$

DEFINITION 1.4.2. Two rays are *asymptotic* if their intersection contains a ray, that is, if they eventually agree. The *boundary* ∂T is the set of asymptotic equivalence classes of rays.

Fix a base vertex $o \in T$.

LEMMA 1.4.3. *For each $\xi \in \partial T$, there is exactly one ray starting at o and representing ξ.*

PROOF. Take a ray $\alpha \in \xi$ and let $\alpha(t_0)$ be the last vertex of α on the segment $[o, \alpha(0)]$. Then $[o, \alpha(t_0)] + \alpha|_{[t_0, \infty)}$ is a ray representing ξ. Uniqueness follows from uniqueness of embedded paths in a tree. □

Extending the Gromov product to ∂T. For $\xi, \eta \in \partial T$, define:

$$(\xi \mid \eta)_o := \text{length of the common initial segment of the rays } \alpha_\xi \in \xi \text{ and } \alpha_\eta \in \eta \text{ based at } o$$

LEMMA 1.4.4. *If $\xi \neq \eta \in \partial T$, there is a unique bi-infinite geodesic (ξ, η) such that each of its two rays from any vertex represents one of ξ or η. Moreover,*

$$(\xi \mid \eta)_o = d(o, (\xi, \eta))$$

THEOREM 1.4.5. *The following defines a metric on ∂T:*

$$d(\xi, \eta) := 2^{-(\xi|\eta)_o}$$

PROOF. Symmetry and nonnegativity are immediate. Also,

$$d(\xi, \eta) = 0 \iff (\xi \mid \eta)_o = \infty \iff \alpha_\xi = \alpha_\eta \iff \xi = \eta$$

For the triangle inequality, the common initial segment of α_ξ and α_ζ has length at least the minimum of the common initial segment lengths for $(\alpha_\xi, \alpha_\eta)$ and $(\alpha_\eta, \alpha_\zeta)$. Thus:

$$(\xi \mid \zeta)_o \geq \min\{(\xi \mid \eta)_o, (\eta \mid \zeta)_o\}$$

This gives the stronger ultrametric inequality:

$$d(\xi, \zeta) \leq \max\{d(\xi, \eta), d(\eta, \zeta)\}$$ □

1.5. Cantor boundaries and compactification

THEOREM 1.5.1. *Let T be a tree in which every vertex has finite valence at least 3. Then ∂T is the Cantor set.*

PROOF. We use the fact that the Cantor set is characterized as the unique nonempty compact metrizable totally disconnected perfect topological space.

A tree without leaves is unbounded, so its boundary is nonempty. We use the metric topology given by Theorem 1.4.5, so metrizability is automatic.

Compactness. Since ∂T is metrizable, compactness is equivalent to sequential compactness. Let (ξ_i) be a sequence in ∂T, and let α_{ξ_i} be the corresponding rays based at o. Only finitely many edges are incident to o, so we can pass to a subsequence whose rays have the same first edge. The opposite vertex to o also has finite valence, so we can pass to a further subsequence with the same first two edges, and continue. A diagonal subsequence converges to the ray α whose first n edges are the stabilized initial n-edge segments. Thus $\xi_{i_k} \to [\alpha]$.

No isolated points (perfectness). Given $\xi \in \partial T$ and $n \geq 1$, take a ray α representing ξ and branch away from it after n edges. Since every vertex has valence at least 3, this produces a distinct ray α_n agreeing with α on the first n edges. Hence $[\alpha_n] \to \xi$.

Totally disconnected. For $\xi \neq \eta$ choose n so that $2^{-n} < d(\xi, \eta)$, and set

$$C_n(\xi) := \{\zeta \in \partial T \mid (\xi \mid \zeta)_o \geq n\}$$

This is the closed ball $\overline{B}_{2^{-n}}(\xi)$, and it is also open. Its complement is open because sufficiently close rays must share the same initial n edges. Thus ξ and η are separated by a clopen set and cannot lie in the same connected component. □

THEOREM 1.5.2 (Compactification). *Let T be a tree of finite valence at least 3. There is a topology on $\bar{T} := T \cup \partial T$ with the following properties:*

(i) $\overline{T}$ is compact.
(ii) Its restrictions to T and ∂T agree with their respective metric topologies.
(iii) T is open and dense in $\overline{T}$.

PROOF. For $x \in T$ use the usual metric balls as a neighborhood basis. For $\xi \in \partial T$, choose a ray α_ξ based at o and take as a neighborhood basis the sets:

$$C_n(\xi) := \{x \in T \mid [o, x] \text{ and } \alpha_\xi \text{ agree for length } \geq n\}$$
$$\cup \{\eta \in \partial T \mid (\xi \mid \eta)_o \geq n\}$$

Each boundary point is represented by a ray, and the vertices of the ray approach the boundary point in this topology, so T is dense. For every point $x \in T$ and every positive r the ball $B_r(x)$ is an open set containing x and contained in T, so T is open.

Compactness is proved by the same diagonal argument as for ∂T. (It turns out that $\bar{T}$ is metrizable, but here it is enough to observe that $\bar{T}$ is second countable, so compact is equivalent to sequentially compact.) Every sequence in ∂T has a convergent subsequence. For a sequence (x_i) in T, either it is bounded, in which case properness of the locally finite tree gives a convergent subsequence, or (after passing to a subsequence) $d(o, x_i) \to \infty$. In the latter case, apply the diagonal argument to the finite segments $[o, x_i]$ to obtain a limiting ray. □

1.6. Isometries of regular trees

Let T_r be a regular tree of valence r, where $3 \le r < \infty$. An *isometry* $\phi\colon T_r \to T_r$ preserves distances:

$$d(x, y) = d(\phi(x), \phi(y)) \qquad (x, y \in T_r)$$

Notice that vertices have valence at least 3, vertices are distinguishable from interior points of edges. Thus, a map $T_r \to T_r$ is an isometry if and only if it is an isomorphism of the tree structure, meaning a bijection of the vertices that preserves adjacency.

An isometry is one of the following types:

Elliptic: It fixes a point of T_r. We will show in this case it fixes a vertex or the midpoint of an edge.

Hyperbolic/loxodromic: it has no fixed point in T_r; it fixes exactly two boundary points ξ^-, ξ^+, translates along the axis (ξ^-, ξ^+), and has north–south dynamics on ∂T_r: for every pair of disjoint neighborhoods $U^- \ni \xi^-$, $U^+ \ni \xi^+$, there exists N such that $\forall n \ge N$:

$$\phi^n(\partial T_r \setminus U^-) \subseteq U^+$$

Also, the axis (ξ^-, ξ^+) is equal to the minimal displacement set of ϕ.

Proof of the classification. If ϕ inverts an edge then it fixes the midpoint of that edge.

Suppose ϕ fixes no vertex and inverts no edge. Suppose u is an interior point of an edge $w \underset{e}{-} x$. Then $\phi(u) \notin e$, so it is an interior point of some different edge $y - z$. Consider the geodesic from u to $\phi(u)$. It goes through either w or x, and it goes through either y or z. Up to relabelling, suppose it goes through x and y. Then $d(x, y) < d(u, \phi(u))$. Now there are two cases: we have $\phi(\{w, x\}) = \{y, z\}$, so either $\phi(x) = y$ or $\phi(x) = z$. In the former case $d(x, \phi(x)) < d(u, \phi(u))$. In the latter case $d(x, \phi(x)) = d(u, \phi(u))$. Thus, for all u there exists a vertex x such that $d(x, \phi(x)) \le d(u, \phi(u))$.

Since vertices move by distances that are nonnegative integers, we have shown that the following quantity is a minimum, and it is either 0 and realized at the midpoint of an edge, or it is realized at a vertex:

$$\tau(\phi) := \inf_{u \in T_r} d(u, \phi(u))$$

The elliptic case is $\tau(\phi) = 0$, which is realized at either the midpoint of an edge or at a vertex.

For the loxodromic case, suppose $\tau(\phi) > 0$ and choose $v \in VT_r$ realizing the positive minimum. We claim that the segments $[v, \phi(v)]$ and $[\phi(v), \phi^2(v)]$ do not overlap in a nontrivial segment. Otherwise, let x be the point of $[v, \phi(v)]$ at distance $1/2$ from $\phi(v)$. Then x is also the point of $[\phi(v), \phi^2(v)]$ at distance $1/2$ from $\phi(v)$, and $\phi(x)$ is the point of $[\phi(v), \phi^2(v)]$ at distance $1/2$ from $\phi^2(v)$, so:

$$d(x, \phi(x)) = d(\phi(v), \phi^2(v)) - 1 = d(v, \phi(v)) - 1$$

This contradicts minimality of $d(v, \phi(v))$.

Consequently the segments

$$[\phi^n(v), \phi^{n+1}(v)] \qquad (n \in \mathbb{Z})$$

concatenate to form a bi-infinite geodesic. The isometry ϕ translates along this axis, and its two endpoints are

$$\xi^+ := \lim_{n\to\infty} \phi^n(v) \qquad \xi^- := \lim_{n\to-\infty} \phi^n(v)$$

If η is different from these endpoints, the geodesic ray from o to $\phi^n(\eta)$ overlaps longer and longer with the ray toward ξ^+, uniformly for η outside any fixed neighborhood of ξ^-. This is the asserted north–south dynamics.

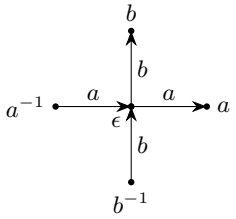
Finally, the axis (ξ^-, ξ^+) is the entire minimal displacement set, as follows. If $u \notin (\xi^-, \xi^+)$ then there is closest point v of (ξ^-, ξ^+) to u. The point v is a vertex, and $[u, v] \cap (\xi^-, \xi^+) = \{v\}$. Apply ϕ: $\phi(v)$ is the closest point of (ξ^-, ξ^+) to $\phi(u)$, $\phi([u, v]) = [\phi(u), \phi(v)]$, and $[\phi(u), \phi(v)] \cap (\xi^-, \xi^+) = \{\phi(v)\}$. Then we claim that v and $\phi(v)$ separate u and $\phi(u)$, so $d(u, \phi(u)) > d(v, \phi(v))$. □

1.7. The free group and its Cayley tree

The group $F_2 = \langle a, b \mid \rangle$ is the free group on two generators. Its elements are uniquely represented by freely reduced words in the letters a, a^{-1}, b, b^{-1}. The identity is the empty word, denoted ϵ. Multiplication is concatenation followed by free reduction. For example,

$$(aba^{-1})(ab^2) = ab^3$$

The group F_2 acts isometrically on the regular tree T_4. The action is free and transitive on the vertices, has no edge inversions, and has two orbits of unoriented edges. We establish these facts by constructing a labelling of the tree. Label one vertex by ϵ. It has 4 incident edges. We declare two of them to be outgoing and label them a and b, and declare the other two to be incoming, and label them a and b. By convention, the label of an edge read against its direction is the inverse of the given label, so from the point of view of ϵ the four outgoing edges are labelled a, b, a^{-1} and b^{-1}. Label the corresponding vertices a, b, a^{-1} and b^{-1}.



Now induct: take a closest vertex to ϵ that is labelled, say by reduced word w, but for which not all of its incident edges are labelled. By construction, exactly one of its incident edges has been directed and labelled, the edge leading to ϵ. Assign directions and labels to the remaining three edges in such a way that vertex w has one incoming and one outgoing edge labelled by each of a and b. Label the adjacent vertices of w as follows. For the outgoing edge labelled a, label the next vertex by the free reduction of wa. For the incoming edge labelled a, label the next vertex by the free reduction of wa^{-1}. Proceed similarly for b. Observe that there is exactly one of these four labels that require free reduction: suppose, for example, that $w = ua$. Then appending a, b, or b^{-1} results in a freely reduced word; only a^{-1} requires a free reduction. Moreover, we arrived at the vertex labelled w from the vertex labelled u by following the outgoing edge at u labelled a, so $wa^{-1} = uaa^{-1} \mapsto u$. Thus, the labels assigned in the induction step are consistent with labels from previous steps.

This process limits to a labelling of the tree in which every vertex is labelled by a freely reduced word of length equal to its distance from ϵ, and every vertex has one incoming and one outgoing edge with each label a and b. In such a labelled tree, an edge path has a label, defined by concatenating the labels of its edges, inverting the label if the edge is traversed against its assigned direction. The label of an embedded edge path is a freely reduced word. Moreover, given a freely reduced word and a vertex, there is a unique edge path starting at that vertex whose label is the given reduced word. It follows that the labelling gives a bijection between vertices of T_4 and elements of F_2, since elements of F_2 are in bijection with freely reduced words,

which are in bijection with embedded edge paths with starting vertex ϵ, thus with vertices of T_4.

Now we claim that left multiplication of F_2 on the labels of the vertices induces an isometry of T_4. This is because edges are of the form $x \xrightarrow{a} xa$, so $w \in F_2$ takes the end vertices of that edge to wx and wxa, which differ by right multiplication of a, so there is an edge $wx \xrightarrow{a} wxa$. Thus, F_2 acts on T_4, freely and transitively on vertices, and preserving the directions and labels of the edges.

Suppose $\epsilon \neq w = uvu^{-1}$ is written as a freely reduced product with v cyclically reduced. Thus there is no cancellation within uvu^{-1}, and the first and last letters of v are not inverses. Multiplication by w has fixed points in ∂T_4:

$$w^{+\infty} = uv^{+\infty} := \lim_{n\to\infty} uv^n \qquad w^{-\infty} = uv^{-\infty} := \lim_{n\to\infty} uv^{-n}$$

These are the two ends of the axis of w.

1.8. Centralizers in the free group

THEOREM 1.8.1. *For $w \in F_2 \setminus \{1\}$, the centralizer*

$$Z_{F_2}(w) := \{x \in F_2 \mid xwx^{-1} = w\}$$

is isomorphic to $\mathbb{Z}$.

PROOF. Let $x \in Z_{F_2}(w)$. The isometry w translates along the bi-infinite axis:

$$A_w = (w^{-\infty}, w^{+\infty})$$

For any $v \in A_w$, we claim that $xv \in A_w$, because A_w is the entire minimal displacement set of the w–action, and:

$$\tau(w) = d(v, wv) = d(xv, xwv) = d(xv, wxv)$$

Therefore $x(A_w) = A_w$. It cannot interchange the two endpoints of A_w, because then it would fix an interior point of the axis, whereas the F_2 action has no nontrivial elliptic elements. Hence x acts on A_w by translation.

Restriction to the axis gives a homomorphism

$$Z_{F_2}(w) \longrightarrow \mathrm{Trans}(A_w) \cong \mathbb{Z}$$

Its kernel is trivial because the action of F_2 on vertices of T_4 is free. The image is a subgroup of $\mathbb{Z}$ and is nontrivial because it contains w. Hence the centralizer is isomorphic to $\mathbb{Z}$. □

1.9. Ping-pong on the boundary

THEOREM 1.9.1 (Ping-pong). *Let $f, g \in F_2 \setminus \{1\}$ satisfy*

$$\left|\{f^{+\infty}, f^{-\infty}, g^{+\infty}, g^{-\infty}\}\right| = 4$$

Then, for some $n \geq 1$, the subgroup generated by f^n and g^n is isomorphic to F_2.

PROOF. Choose pairwise disjoint neighborhoods

$$A^+ \ni f^{+\infty}, \quad A^- \ni f^{-\infty}, \quad B^+ \ni g^{+\infty}, \quad B^- \ni g^{-\infty}$$

which do not together cover all of ∂T_4. By north–south dynamics, choose n large enough that

$$f^n(\partial T_4 \setminus A^-) \subseteq A^+$$
$$f^{-n}(\partial T_4 \setminus A^+) \subseteq A^-$$
$$g^n(\partial T_4 \setminus B^-) \subseteq B^+$$
$$g^{-n}(\partial T_4 \setminus B^+) \subseteq B^-$$

Set $\alpha = f^n$ and $\beta = g^n$. We wish to prove that every nonempty freely reduced word w in $\alpha^{\pm 1}, \beta^{\pm 1}$ is nontrivial. Choose $\xi \in \partial T_4$ outside all four sets A^+, A^-, B^+, B^-. It suffices to show that $w\xi \neq \xi$.

The point ξ is the "ping-pong ball", and the game is that the letters of w hit the ball into their designated regions: If, for example, the last letter of w is α, so $w = w'\alpha$, then the first player is α, and, since $\xi \notin A^-$, α hits ξ into A^+, ie, $\alpha\xi \in A^+$. The next player is the last letter of w'. Since w is freely reduced, the next player is not α^{-1}, it is one of α, β, or β^{-1}. Since the ball currently lies in A^+, it is not in A^-, B^-, or B^+, so player 2 hits the ball into A^+, B^+, or B^-, respectively. Continue, inducting on the length of w, to see that from the first hit onwards, the ball is always in one of the sets A^+, A^-, B^+, or B^-, so it never returns to the starting point ξ. Thus, $w\xi \neq \xi$, so $w \neq 1$. □

1.10. Exercises

EXERCISE 1.1. Prove Lemma 1.2.1.

EXERCISE 1.2. Prove Lemma 1.3.5.

EXERCISE 1.3. Prove Lemma 1.4.4.

EXERCISE 1.4. **Gromov products and the four-point condition.**
Let T be a tree and $o \in T$.

(a) Show that for every $x, y, z \in T$:
$$(x \mid z)_o \geq \min\{(x \mid y)_o, (y \mid z)_o\}$$

(b) Show that for any four points $x, y, z, w \in T$, the two largest of the following quantities are equal:
$$d(x, y) + d(z, w)$$
$$d(x, z) + d(y, w)$$
$$d(x, w) + d(y, z)$$

(c) Interpret this exercise as a statement about geodesic quadrilaterals in a tree.

EXERCISE 1.5. **Nearest-point projections.**
Let C be a nonempty connected subgraph of a tree T.

(a) Show C is a tree.
(b) Show that every $x \in T$ has a unique closest point $\pi_C(x) \in C$.
(c) Show that for every $y \in C$, the geodesic $[x, y]$ passes through $\pi_C(x)$.
(d) Show that $\pi_C \colon T \to C$ is 1-Lipschitz. A map ϕ is *L–Lipschitz* if $d(\phi(x), \phi(y)) \leq Ld(x, y)$ for all x and y.

EXERCISE 1.6. **The Morse lemma for trees.**
A (not necessarily continuous!) map $\gamma \colon I \to X$ is a (λ, ε)-*quasigeodesic* if $\lambda \geq 1$, $\varepsilon \geq 0$, and for all s, t in the interval I:
$$\frac{1}{\lambda}|s - t| - \varepsilon \leq d(\gamma(s), \gamma(t)) \leq \lambda|s - t| + \varepsilon$$
Suppose γ is a continuous (λ, ε)-quasigeodesic from x to y in a tree T.

(a) Use Lemma 1.3.6 to show that $[x, y]$ is contained in the image of γ.
(b) For $z \in \gamma([a, b]) \setminus [x, y]$, let p be its closest point on $[x, y]$. Show that there exist $s < t < u$ such that $\gamma(s) = \gamma(u) = p$ and $\gamma(t) = z$.
(c) Deduce that:
$$d(z, [x, y]) \leq \frac{\lambda^2\varepsilon}{2} + \varepsilon$$
(d) Show that every $(\lambda, 0)$-quasigeodesic in a tree has the same image as the geodesic between its endpoints.

EXERCISE 1.7. **Changing the basepoint.**
Let T be a tree and let $o, o' \in VT$.

(a) Show that for $\xi, \eta \in \partial T$:
$$|(\xi \mid \eta)_o - (\xi \mid \eta)_{o'}| \leq d(o, o')$$

(b) Deduce that the two metrics given by Theorem 1.4.5 satisfy:

$$2^{-d(o,o')} d_o(\xi,\eta) \leq d_{o'}(\xi,\eta) \leq 2^{d(o,o')} d_o(\xi,\eta)$$

(c) Conclude that the topology of ∂T is independent of the choice of basepoint.

EXERCISE 1.8. **Translation lengths.**
Let g be a hyperbolic isometry of T_r, with axis A_g.

(a) Prove that for every $n \in \mathbb{Z}$:

$$\tau(g^n) = |n|\tau(g)$$

(b) Prove that for every $x \in T_r$ and every positive integer n:

$$d(x, g^n x) = n\tau(g) + 2d(x, A_g)$$

(c) Deduce that:

$$\lim_{n\to\infty} \frac{d(x, g^n x)}{n} = \tau(g)$$

EXERCISE 1.9. **North–south dynamics.**
Give a detailed proof of the north–south dynamics assertion from the classification of isometries of T_r.

EXERCISE 1.10. **Common boundary fixed points.**
Let $f, g \in F_2 \setminus \{1\}$.

(a) Suppose that f and g have a common fixed point in ∂T_4. Show that their axes have a common infinite ray.
(b) Show that the commutator $[f, g]$ fixes a vertex of T_4.
(c) Deduce that f and g commute.
(d) Conclude that f and g belong to a common cyclic subgroup of F_2, and that they have the same two fixed points in ∂T_4.
(e) Deduce from Theorem 1.9.1 that if f and g do not belong to a common cyclic subgroup, then for sufficiently large n:

$$\langle f^n, g^n \rangle \cong F_2$$

LECTURE 2

The hyperbolic plane and surface groups

2.1. The Poincaré disk and its metric

In the first lecture we had a metric space definition of the length of a continuous path. Recall from calculus that the length of a C^1 curve $\gamma\colon [a,b] \to \mathbb{R}^n$ is:

$$\int_a^b |\gamma'(t)|\, dt$$

The Euclidean length of $\gamma'(t)$ is measured using the usual inner product. We can instead choose a different inner product for tangent vectors at each point.

Definition 2.1.1 (Poincaré disk). Let $\mathbb{D} := \{z \in \mathbb{C} \mid |z| < 1\}$. At $z \in \mathbb{D}$ define an inner product on tangent vectors $v, w \in T_z\mathbb{D} \cong \mathbb{C}$ by:

$$\langle v, w\rangle_z := \frac{4\,\mathrm{Re}(v\bar{w})}{(1-|z|^2)^2} = \frac{4\langle v, w\rangle_{\mathbb{E}^2}}{(1-|z|^2)^2}$$

Thus the length of a C^1 curve $\gamma\colon [a,b] \to \mathbb{D}$ is:

$$L_{\mathbb{H}}(\gamma) := \int_a^b \frac{2|\gamma'(t)|}{1-|\gamma(t)|^2}\, dt$$

Define a metric on $\mathbb{D}$ by:

$$d_{\mathbb{H}}(z,w) := \inf\{L_{\mathbb{H}}(\gamma) \mid \gamma \text{ is piecewise } C^1 \text{ curve joining } z \text{ to } w\}$$

The resulting length space is the *hyperbolic plane* $\mathbb{H}^2$. This specific description of the hyperbolic plane is the *Poincaré disk model*.

The factor 4 normalizes the sectional curvature to -1. For us, the important thing is that the curvature is negative, not what its magnitude is. For comparison, the Euclidean plane has sectional curvature 0, and the unit sphere with its usual metric has sectional curvature $+1$.

Notice that at every point z the inner product on $T_z\mathbb{D}$ is a rescaling of the Euclidean inner product. The lengths of the vectors are different, but angles between them are the same as in the Euclidean sense. In a small neighborhood around z all tangent planes are scaled by close factors, so on small neighborhoods the metric behaves much like the Euclidean metric.

Example 2.1.2. For $a < 1$, let $\gamma\colon [0,a] \to \mathbb{H}^2 : t \mapsto t$. We have $\gamma'(t) = 1$ for all t. In Euclidean terms γ has constant unit speed.

$$\begin{aligned} L_{\mathbb{H}^2}(\gamma) &= \int_0^a |\gamma'(t)|_{T_{\gamma(t)}\mathbb{H}^2} dt \\ &= \int_0^a \frac{2|\gamma'(t)|_{\mathbb{E}^2}}{1-|\gamma(t)|^2} dt \\ &= 2\int_0^a \frac{1}{1-t^2} dt \\ &= \log\frac{1+t}{1-t}\Big|_0^a \\ &= \log\frac{1+a}{1-a} \end{aligned}$$

Notice as a gets close to 1, this curve becomes arbitrarily long.

We cannot determine the distance from 0 to a from this example alone. We have only computed the length of one curve. Potentially there could be some shorter curve with the same endpoints. It turns out that that is not the case.

PROPOSITION 2.1.3. *Projection to any line through* 0 *in* $\mathbb{H}^2$ *is length non-increasing.*

PROOF. Let π be orthogonal projection to a line L. Choose a unit vector in the direction of L and extend it to an orthonormal basis of $\mathbb{R}^2$. Let γ be any differentiable curve. The tangent vector to $\pi \circ \gamma$ at t in the new coordinates is the same as $\gamma'(t)$, but with the second coordinate set to 0, so its Euclidean length does not increase. Furthermore, $\pi \circ \gamma(t)$ is no farther from the origin than $\gamma(t)$, so $|\pi \circ \gamma|_{\mathbb{H}^2} \leq |\gamma|_{\mathbb{H}^2}$. □

PROPOSITION 2.1.4 (Radial distance and geodesics). *For* $z \in \mathbb{D}$*:*

$$d_{\mathbb{H}}(0, z) = 2 \int_0^{|z|} \frac{dr}{1 - r^2} = \log \frac{1 + |z|}{1 - |z|}$$

The radial segment from 0 *to* z *is the unique geodesic between its endpoints, up to reparametrization.*

PROOF. The displayed integral gives the length of the radial segment. Every curve from 0 to z projects onto a curve in the diameter through z with no greater length. This projected curve must traverse the radial segment and hence has length at least the displayed quantity. Equality is possible only if the original curve stays on that diameter and has no backtracking. □

In particular, points approaching the boundary circle are infinitely far from the origin:

$$d_{\mathbb{H}}(0, z) \longrightarrow \infty \qquad \text{as } |z| \longrightarrow 1$$

Recall:

$$\sinh(x) := \frac{e^x - e^{-x}}{2} \qquad \cosh(x) := \frac{e^x + e^{-x}}{2} \qquad \tanh(x) := \frac{\sinh(x)}{\cosh(x)} = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

LEMMA 2.1.5. *For* $z \in \mathbb{D}$*:*

$$d_{\mathbb{H}^2}(0, z) = 2 \operatorname{artanh}(|z|)$$

Equivalently:

$$d_{\mathbb{H}^2}(0, z) = 2 \operatorname{arsinh}\left(\frac{|z|}{\sqrt{1 - |z|^2}}\right)$$

PROOF. A direct computation using Proposition 2.1.4 yields the first. The second follows from the identity $\sinh(\operatorname{artanh}(r)) = \frac{r}{\sqrt{1-r^2}}$. □

2.2. The isometry group

We take the following theorem from Riemannian geometry as a black box:

THEOREM 2.2.1 (Myers–Steenrod). *A distance-preserving bijection between connected Riemannian manifolds is smooth. Moreover, an isometry is uniquely determined by its value and its derivative at one point.*

Rotations $z \mapsto e^{i\theta} z$ and complex conjugation $z \mapsto \bar{z}$ clearly preserve the hyperbolic metric: they preserve Euclidean lengths and distances from 0. But all of those maps fix 0.

PROPOSITION 2.2.2. *Consider:*

$$\mathrm{PSU}(1,1) := \left\{ \left[\begin{pmatrix} a & b \\ \bar{b} & \bar{a} \end{pmatrix} \right] \mid \begin{pmatrix} a & b \\ \bar{b} & \bar{a} \end{pmatrix} \in \mathcal{M}_{2\times 2}(\mathbb{C}),\ |a|^2 - |b|^2 = 1 \right\}$$

(1) $\mathrm{PSU}(1,1)$ *acts on* $\mathbb{H}^2$*.*
(2) $\mathrm{PSU}(1,1)$ *contains* $\mathrm{SO}(2) < \mathrm{Isom}^+(\mathbb{H}^2)$*.*
(3) $\mathrm{PSU}(1,1)$ *contains a 1-parameter family that acts by hyperbolic translation on the real axis.*
(4) $\mathrm{PSU}(1,1)$ *acts transitively on the set of unit tangent vectors to* $\mathbb{H}^2$*.*

(5) $\mathrm{Isom}^+(\mathbb{H}^2)$ *acts simply transitively on the set of unit tangent vectors to* $\mathbb{H}^2$*. Consequently,* $\mathrm{Isom}^+(\mathbb{H}^2) \cong \mathrm{PSU}(1,1)$.

PROOF. Let $\mathrm{SU}(1,1) := \{ \begin{pmatrix} a & b \\ \bar{b} & \bar{a} \end{pmatrix} \in \mathcal{M}_{2\times 2}(\mathbb{C}) \mid |a|^2 - |b|^2 = 1\}$ act on $\mathbb{C} \cup \{\infty\}$ by Möbius transformations, so that:

$$\begin{pmatrix} a & b \\ \bar{b} & \bar{a} \end{pmatrix} : \ z \mapsto \frac{az+b}{\bar{b}z + \bar{a}}$$

Composition of Möbius transforms is compatible with matrix multiplication, so such a map is a bijection of $\mathbb{C} \cup \{\infty\}$. Observe that the center of $\mathrm{SU}(1,1)$ consists of plus and minus the identity matrix, and this is also the kernel of the action. $\mathrm{PSU}(1,1) := \mathrm{SU}(1,1)/\pm \mathrm{Id}$.

For $c \in \mathbb{D}$, let $F_c(z) := \frac{z+c}{\bar{c}z+1}$. Note that if $|a|^2 - |b|^2 = 1$ then $a \neq 0$ and $|b/a|^2 = \frac{|b|^2}{1+|b|^2} < 1$. Then:

$$\frac{az+b}{\bar{b}z+\bar{a}} = \frac{a}{\bar{a}} \frac{z + b/a}{(\bar{b}/\bar{a})z + 1} = e^{i\theta} F_c(z) \quad \text{for} \quad \theta := \arg\left((a/|a|)^2\right) \text{ and } c := b/a$$

$F_c(z)$ sends 0 to c and has inverse $F_{-c}(z)$.

$$1 - |F_c(z)|^2 = \frac{(1-|c|^2)(1-|z|^2)}{|1+\bar{c}z|^2} > 0$$

F_c is holomorphic on $\mathbb{D}$, so the derivative $D_z F_c \colon T_z\mathbb{D} \to T_{F_c(z)}\mathbb{D}$ is given by complex multiplication by $F_c'(z)$.

$$|F_c'(z)| = \left| \frac{1(\bar{c}z+1) - \bar{c}(z+c)}{(\bar{c}z+1)^2} \right| = \frac{1-|c|^2}{|\bar{c}z+1|^2}$$

Consequently, for $v \in T_z\mathbb{D}$:

$$\sqrt{\langle D_z F_c(v), D_z F_c(v)\rangle_{F_c(z)}} = \frac{2|F_c'(z)v|}{1-|F_c(z)|^2} = \frac{2|v|}{1-|z|^2} = \sqrt{\langle v, v\rangle_z}$$

So F_c preserves the hyperbolic length of tangent vectors, so it preserves hyperbolic lengths of curves, hence the hyperbolic metric.

A copy of $SO(2)$ fixing 0 is obtained by taking $b = c = 0$.

Translations along the real axis are obtained by taking $\theta = 0$ and real c, which is equivalent to a and b being real. For translation by hyperbolic distance t, take $c := \tanh(t/2)$.

For any $z \in \mathbb{D}$ and $v \in T_z\mathbb{D}$, the isometry F_{-z} takes z to 0 and its derivative takes v to a vector in $F_{-z}'v \in T_0\mathbb{D}$ of the same hyperbolic norm as v. If $\theta := -\arg(F_{-z}'v)$ then $e^{i\theta}F_{-z}$ is an isometry taking z to 0 and taking v to the unique vector in $T_0\mathbb{D}$ pointing in the positive real direction and with hyperbolic norm equal to that of v.

Finally, we claim that if $\phi \in \mathrm{Isom}^+(\mathbb{H}^2)$ fixes a nonzero tangent vector then $\phi = \mathrm{Id}_{\mathbb{H}^2}$. By the previous paragraph, we may assume the fixed vector is in $T_0\mathbb{D}$. At $z = 0$ the hyperbolic norm is a scaling of the Euclidean norm, so the group preserving the norm and the orientation is $\mathrm{SO}(2)$, which is rotations, so if some vector is fixed then the rotation is trivial. We know an isometry, $\mathrm{Id}_{\mathbb{H}^2}$ that fixes 0 and whose derivative at 0 is the identity on $T_0\mathbb{D}$. By Theorem 2.2.1, $\phi = \mathrm{Id}_{\mathbb{H}^2}$. □

COROLLARY 2.2.3 (Distance formula). *For* $z, w \in \mathbb{D}$*:*

$$d_{\mathbb{H}}(z,w) = 2\,\mathrm{artanh}\left|\frac{z-w}{1-\bar{w}z}\right| = 2\,\mathrm{arsinh}\left(\frac{|z-w|}{\sqrt{(1-|z|^2)(1-|w|^2)}}\right)$$

PROOF. The isometry F_{-w} takes w to 0. Apply Lemma 2.1.5 to $F_{-w}(z)$, using

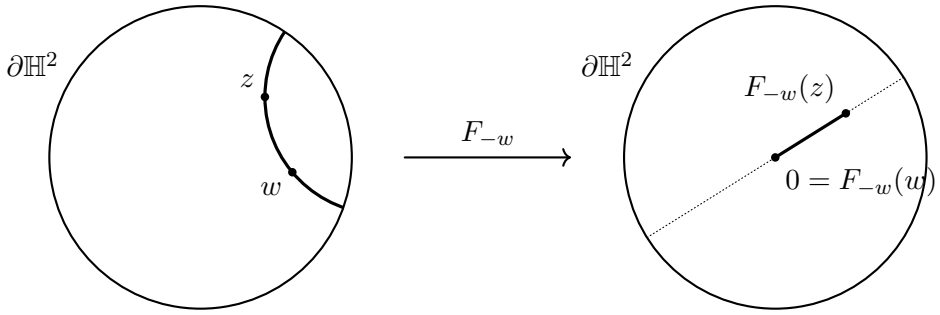
$$1 - \left|\frac{z-w}{1-\bar{w}z}\right|^2 = \frac{(1-|z|^2)(1-|w|^2)}{|1-\bar{w}z|^2}$$

□

2.3. Geodesics and the circle at infinity

PROPOSITION 2.3.1 (Geodesics in the disk). *Between any two points of $\mathbb{H}^2$ there is a unique geodesic, up to reparametrization. Every geodesic segment extends uniquely to a bi-infinite geodesic. Its image in the disk is either an arc of a Euclidean circle orthogonal to S^1 or a Euclidean diameter (which may be thought of as an arc of a "generalized circle", Euclidean circle of infinite radius, orthogonal to S^1).*

PROOF. Given $z, w \in \mathbb{D}$, move w to 0 by an isometry F_{-w}, take the radial line from 0 to $F_{-w}(z)$, which is a geodesic segment, by Proposition 2.1.4, then move it back with F_w. This also gives uniqueness and extension to a complete line. Möbius maps send generalized circles to generalized circles, and are conformal: they preserve angles. Their images of diameters are therefore generalized circles that meet the unit circle orthogonally. □



The circle S^1 is not part of the metric space: it is infinitely far away. But every geodesic ray converges to a point of S^1. As in Lecture 1, we think of these points as *directions at infinity.*

We call two geodesic rays *asymptotic* if their images have finite Hausdorff distance. In a tree, this agrees with the definition from Lecture 1: two rays are asymptotic if they eventually coincide.

PROPOSITION 2.3.2 (Boundary of the hyperbolic plane). *Two geodesic rays in $\mathbb{H}^2$ are asymptotic if and only if their limiting endpoints on S^1 agree. Thus:*

$$\partial\mathbb{H}^2 \cong S^1$$

Every pair of distinct points of $\partial\mathbb{H}^2$ is joined by a unique bi-infinite geodesic.

PROOF. Suppose that geodesic rays α and β have the same endpoint in S^1. Up to isometry, we may assume that $\alpha = [0, 1)$. If β lies on the real diameter, the conclusion is immediate. Otherwise, β is a subray of a Euclidean circle with center $(1, c)$ for some $c \neq 0$.

The circle containing β has equation:

$$(x-1)^2 + y^2 = 2cy$$

For $z = x + iy \in \beta$ approaching 1, this gives $y = O((1-x)^2)$. By Corollary 2.2.3:

$$\sinh\left(\frac{d_{\mathbb{H}}(z,x)}{2}\right) = \frac{|y|}{\sqrt{(1-x^2-y^2)(1-x^2)}} = O(1-x) \longrightarrow 0$$

Thus the two rays approach one another, and in particular have finite Hausdorff distance.

Conversely, suppose α and β have finite Hausdorff distance, say at most R. Choose $z_n \in \alpha$ converging to its endpoint $\xi \in S^1$, and $w_n \in \beta$ with $d_{\mathbb{H}}(z_n, w_n) \leq R$. By Corollary 2.2.3:

$$|z_n - w_n| \leq 2\sinh(R/2)\sqrt{(1-|z_n|^2)(1-|w_n|^2)} \longrightarrow 0$$

Hence $w_n \to \xi$, so β has endpoint ξ as well.

Finally, two distinct points of S^1 determine a unique generalized Euclidean circle orthogonal to S^1, which gives the unique bi-infinite geodesic joining them. □

2.4. Classification of isometries

THEOREM 2.4.1 (Classification of orientation-preserving isometries). *Every nonidentity orientation-preserving isometry of $\mathbb{H}^2$ is exactly one of the following:*

Elliptic: *It fixes a unique point in $\mathbb{H}^2$ and no points in $\partial\mathbb{H}^2$. It is conjugate to a nontrivial rotation $z \mapsto e^{i\theta}z$ of the disk, $\theta \in \mathbb{R} \setminus \{0\}$.*

Parabolic: *It has no fixed point in $\mathbb{H}^2$ and exactly one fixed point in $\partial\mathbb{H}^2$. It is conjugate to*

$$p_s(z) = \frac{(1+is)z - is}{isz + 1 - is} \qquad (s \in \mathbb{R} \setminus \{0\})$$

which fixes $1 \in S^1$.

Hyperbolic: *It has no fixed point in $\mathbb{H}^2$ and exactly two fixed points in $\partial\mathbb{H}^2$. It is conjugate to*

$$h_t(z) = F_{\tanh(t/2)}(z) = \frac{z + \tanh(t/2)}{1 + z\tanh(t/2)} \qquad (t \in \mathbb{R} \setminus \{0\})$$

which fixes -1 and 1 and translates along the real diameter by signed distance t.

PROOF. By Proposition 2.2.2, every orientation-preserving isometry ϕ is represented by a matrix

$$A = \begin{pmatrix} a & b \\ \bar{b} & \bar{a} \end{pmatrix} \in SU(1,1)$$

The fixed points of the corresponding Mobius transformation satisfy:

$$\bar{b}z^2 + (\bar{a} - a)z - b = 0 \tag{1}$$

The discriminant is:

$$\Delta = (\bar{a} - a)^2 + 4|b|^2 = 4((\operatorname{Re} a)^2 - 1) \in \mathbb{R}$$

A key observation is that $\phi(z) = z$ then $\phi(1/\bar{z}) = 1/\bar{z}$, so if ϕ fixes a point z with $|z| \neq 1$ then the second fixed point is $1/\bar{z} \neq z$, and one of these points has norm strictly less than one, while the other has norm strictly greater than one. Thus ϕ has exactly one fixed point in $\mathbb{H}^2$ and none in $\partial\mathbb{H}^2$. Up to conjugation, we may assume the fixed point in $\mathbb{H}^2$ is 0, so ϕ is a rotation. This is the $\Delta < 0$ case.

If $\Delta \geq 0$ then there are either distinct solutions on $\mathbb{S}^1 = \partial\mathbb{H}^2$, the $\Delta > 0$ case, or a repeated solution on $\mathbb{S}^1$, the $\Delta = 0$ case. In the first case, up to conjugation we may assume that the two fixed points are ± 1. Then the isometry fixes the unique hyperbolic geodesic that has ± 1 as its limit points, so it must be translation along that axis.

In the repeated root case we may assume up to conjugation that the unique fixed point is 1. Since $\Delta = 0$, we may replace A by $-A$ if necessary and write $a = 1 + is$ for some real s. The conditions $|a|^2 - |b|^2 = 1$ and that 1 is fixed imply $b = -is$. Thus the transformation is p_s. □

LEMMA 2.4.2 (Translation length and classification). *For an isometry ϕ of a metric space X, define its* translation length *by:*

$$\tau(\phi) := \inf_{x \in X} d(x, \phi(x))$$

Let ϕ be a nonidentity orientation-preserving isometry of $\mathbb{H}^2$. Then:

1. *ϕ is elliptic if and only if $\tau(\phi) = 0$ and the infimum is attained.*
2. *ϕ is parabolic if and only if $\tau(\phi) = 0$ and the infimum is not attained.*
3. *ϕ is loxodromic (hyperbolic) if and only if $\tau(\phi) > 0$. In this case the infimum is attained precisely on the axis of ϕ.*

PROOF. Translation length is invariant under conjugation, so by Theorem 2.4.1 it suffices to consider the standard representatives.

An elliptic isometry fixes a point, so its translation length is 0 and the infimum is attained.

For the parabolic isometry p_s, the distance formula gives:

$$\sinh\left(\frac{d_{\mathbb{H}}(z, p_s z)}{2}\right) = \frac{|s||1 - z|^2}{1 - |z|^2}$$

In particular, for $0 \leq r < 1$:

$$\sinh\left(\frac{d_{\mathbb{H}}(r, p_s(r))}{2}\right) = |s|\frac{1-r}{1+r} \longrightarrow 0$$

as $r \to 1$. Thus $\tau(p_s) = 0$, but the infimum is not attained since p_s has no fixed point in $\mathbb{H}^2$.

Finally, consider the hyperbolic isometry $h_t = F_c$, where $c = \tanh(t/2) \in (-1, 1) \setminus \{0\}$. By the distance formula:

$$\sinh\left(\frac{d_{\mathbb{H}}(z, h_t z)}{2}\right) = \frac{|c||1-z^2|}{\sqrt{1-c^2}(1-|z|^2)} \geq \frac{|c|}{\sqrt{1-c^2}} = \sinh\left(\frac{|t|}{2}\right)$$

Equality holds precisely when z lies on the real diameter, which is the axis of h_t. Thus $\tau(h_t) = |t| > 0$. □

LEMMA 2.4.3 (Trajectories of the standardized 1-parameter isometry families).

- *The 1–parameter family of elliptic isometries $z \mapsto e^{i\theta}z$ fixing 0 has trajectories that are Euclidean circles centered at 0.*
- *The 1–parameter family of loxodromic isometries $h_t(z)$ fixing ± 1 have trajectories that are "hypercycles": Euclidean spheres that intersect $\mathbb{S}^1$, at ± 1.*
- *The 1–parameter family of parabolic isometries $p_s(z)$ fixing 1 have trajectories that are "horocycles": Euclidean spheres tangent to $\mathbb{S}^1$, at 1.*

PROOF OF THE PARABOLIC CASE.

$$\begin{aligned} 1 - p_s(z) &= \frac{1-z}{isz_1 - is} \\ \frac{1}{1-p_s(z)} &= \frac{isz + 1 - is}{1-z} \\ &= \frac{is(z-1)+1}{1-z} \\ &= \frac{1}{1-z} - is \end{aligned}$$

Observe that as $s \in \mathbb{R}$ varies, only the imaginary part of the right-hand side changes, the real part is constant. Let:

$$c := \mathrm{Re}(\frac{1}{1-p_s(z)}) = \mathrm{Re}(\frac{1}{1-z})$$

Set $z = x + iy$, so that:

$$c = \frac{1-x}{(1-x)^2 + y^2} > 0$$

Solve:

$$\left(x - (1 - \frac{1}{2c})\right)^2 + y^2 = \left(\frac{1}{2c}\right)^2$$

That is the equation of a Euclidean circle of radius $\frac{1}{2c}$ centered at $1 - \frac{1}{2c}$, so tangent to $\mathbb{S}^1$ at 1. Thus, for any given $z \in \mathbb{H}^2$ its trajectory under the maps p_s for $s \in \mathbb{R}$ vary along a fixed horocycle, according to the corresponding c value. Moreover, it is easy to check that as $s \to \pm\infty$, $p_s(z)$ goes to 1, so it fills the entire horocycle. □

LEMMA 2.4.4 (North–south dynamics). *Let ϕ be a loxodromic isometry of $\mathbb{H}^2$. Then ϕ has two fixed points $\xi^+, \xi^- \in \partial\mathbb{H}^2$ such that:*

$$\phi^n(\xi) \longrightarrow \xi^+ \qquad \text{for all } \xi \in \partial\mathbb{H}^2 \setminus \{\xi^-\}$$

and:

$$\phi^{-n}(\xi) \longrightarrow \xi^- \qquad \text{for all } \xi \in \partial\mathbb{H}^2 \setminus \{\xi^+\}$$

Both convergences are uniform on compact subsets of the indicated domains.

PROOF. By Theorem 2.4.1, up to conjugation we may assume $\phi = h_t$ for some $t > 0$. Its boundary fixed points are $\xi^+ = 1$ and $\xi^- = -1$.

Recall that:

$$h_t^n = F_{\tanh(nt/2)}$$

Writing $c_n = \tanh(nt/2)$, we have $c_n \to 1$ and:

$$h_t^n(z) - 1 = \frac{(1 - c_n)(z - 1)}{1 + c_n z}$$

This tends to 0 uniformly on compact subsets of $S^1 \setminus \{-1\}$. Thus $h_t^n(\xi) \to 1$ uniformly on such subsets.

Similarly, $h_t^{-n} = F_{-c_n}$, and:

$$h_t^{-n}(z) + 1 = \frac{(1 - c_n)(z + 1)}{1 - c_n z}$$

which tends to 0 uniformly on compact subsets of $S^1 \setminus \{1\}$. □

Orientation-reversing isometries behave differently. For example, complex conjugation $z \mapsto \bar{z}$ is reflection in the real diameter: it fixes that geodesic pointwise and exchanges its two sides. Conjugating by isometries gives reflections in arbitrary geodesics.

Composing reflection in the real diameter with a nontrivial translation along that diameter gives a *glide reflection*:

$$g_t(z) = h_t(\bar{z}) \qquad (t \neq 0)$$

It preserves the real diameter and translates along it, but has no fixed point in $\mathbb{H}^2$. Moreover, $g_t^2 = h_{2t}$. Every orientation-reversing isometry is either a reflection or a glide reflection, Exercise 2.2.

2.5. Hyperbolic surfaces and surface groups

PROPOSITION 2.5.1 (Regular hyperbolic polygons). *For every $n \geq 3$ and every angle*

$$0 < \alpha < \frac{n-2}{n}\pi$$

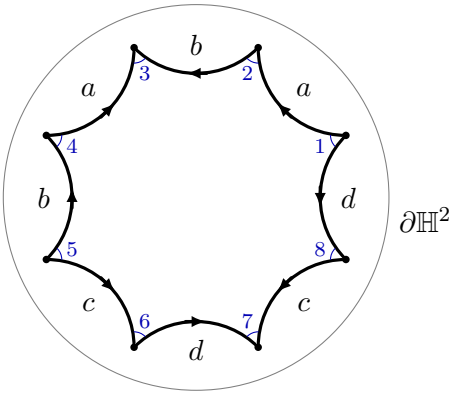
there exists a regular geodesic n–gon in $\mathbb{H}^2$ whose interior angles are all α.

IDEA OF THE PROOF. Place n points at equal angular intervals on a Euclidean circle of radius $r < 1$ centered at 0, and join consecutive points by hyperbolic geodesics. The resulting polygon is regular by rotational symmetry. Its interior angle depends continuously on r. As $r \to 0$ the angle tends to the Euclidean value $(n-2)\pi/n$; as $r \to 1$ it tends to 0. The intermediate value theorem completes the argument. □

Choose a regular hyperbolic octagon P with interior angles $\pi/4$. Label its oriented sides, in cyclic order, by the word:

$$aba^{-1}b^{-1}cdc^{-1}d^{-1}$$

Join each side to the side bearing its inverse label, reversing boundary orientations. The label tells us which two sides are paired; the pairing is realized by an orientation-preserving hyperbolic isometry.

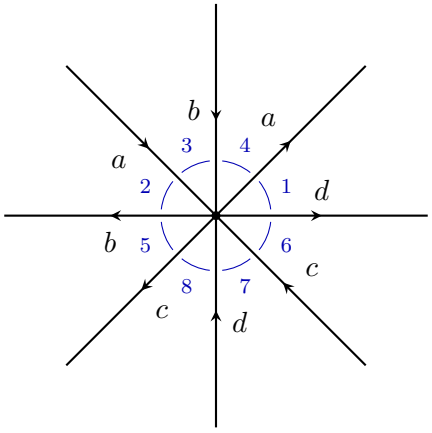


PROPOSITION 2.5.2 (The octagon surface). *The quotient obtained by identifying the sides of P is a closed, orientable surface S_2 of genus two. Moreover, the side-pairing isometries generate a group $\Gamma_2 \leq \mathrm{Isom}^+(\mathbb{H}^2)$ with P as a fundamental polygon and:*

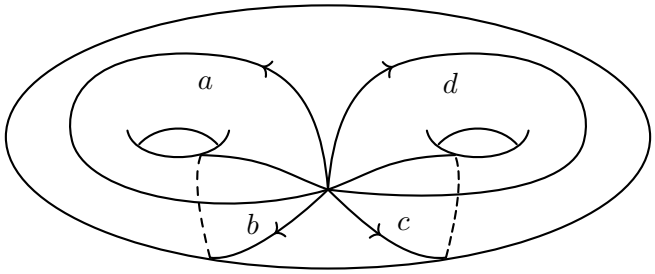
$$\mathbb{H}^2/\Gamma_2 \cong S_2$$

The action is free, properly discontinuous, and cocompact.

PROOF SKETCH. Let S be the quotient obtained from P by the given side identifications, viewed as a 2–complex. First check that all eight vertices of P become a single vertex in S. The 1–skeleton of S is a rose with four petals, directed and labelled by a, b, c, and d, with directions and labels determining the attaching map of the unique 2–cell. Since each label occurs twice on ∂P, with opposite orientations, each edge has two incident sides of the 2–cell. Check that the link of the unique vertex is a circle by observing that P has 8 numbered corners, each with two incident directed, labelled edges, which after gluing fit together cyclically as shown here:



This makes S a closed orientable surface with Euler characteristic $1 - 4 + 1 = -2$, so S has genus 2.



The hyperbolic metric on P defines a hyperbolic metric on S, as follows. For each point $p \in S$ we describe a small neighborhood and define a metric on it.

If p lies in the interior of the 2–cell, it has a unique lift $\tilde{p}$ to P. Choose $\epsilon > 0$ small enough that the ϵ–ball around $\tilde{p}$ is contained in the interior of P. Its image in S is an open neighborhood of p, which we equip with the metric transported from the ball around $\tilde{p}$.

If p lies in the interior of an edge, it has two lifts $\tilde{p}_1$ and $\tilde{p}_2$ to P, one on each of the paired sides. Take ϵ small enough that the ϵ–balls around $\tilde{p}_1$ and $\tilde{p}_2$ intersect ∂P in a simple arc containing the respective lifted point. Then the intersections of these ϵ-balls with P are hyperbolic half-balls, since the sides of P are geodesic. The side identification glues these half–balls along their bounding diameters. We equip their union with the metric obtained by this gluing, making it a hyperbolic ϵ–ball.

Finally, let p be the unique vertex of S. Its eight lifts are the vertices of P. For sufficiently small ϵ, the intersection of P with the ϵ–ball around each lift is a hyperbolic sector of angle $\pi/4$. The link computation shows that these eight sectors are glued in a cycle. We equip their union with the metric obtained by gluing the sectors isometrically along their radial edges. Since their angles sum to 2π, the result is a hyperbolic ϵ–ball.

These local metrics agree on overlaps, since they are all obtained from the hyperbolic metric on P using the given isometric side identifications. Thus they define a hyperbolic metric on S.

Since S is compact, this metric is complete. The universal cover $\widetilde{S}$, with its lifted metric, is therefore a complete simply connected surface that is locally isometric to $\mathbb{H}^2$. Then it is a theorem that it actually is isometric to $\mathbb{H}^2$: there is only one complete, simply connected surface of constant curvature -1.

The map $P \to S$ lifts to a map $P \to \widetilde{S} \cong \mathbb{H}^2$. On the interior of P, this is a local isometry, hence the restriction of a hyperbolic isometry of $\mathbb{H}^2$. Thus P lifts to an isometrically embedded octagon. Choose the isometry $\widetilde{S} \cong \mathbb{H}^2$ so that this lifted octagon is identified with P itself.

The interior of the unique 2–cell is $S \setminus S^{(1)}$, which is simply connected. Consequently, each component of its preimage in $\widetilde{S}$ maps homeomorphically onto it. The closures of these components are the translates of P under the deck transformation group, and give a tiling of $\mathbb{H}^2$. The deck transformation group acts simply transitively on these copies of P. The transformations taking P to its neighbors across paired sides are the prescribed side-pairing isometries or their inverses. Since any two copies can be joined by a sequence of adjacent copies, these transformations generate the deck group.

Denote this deck transformation group by $\Gamma_2 \leq \mathrm{Isom}^+(\mathbb{H}^2)$. Then P is a fundamental polygon for Γ_2, and $\mathbb{H}^2/\Gamma_2 \cong S_2$. The action is free and properly discontinuous because it is a deck group action, and cocompact because S_2 is compact.

Reading the boundary word of the octagon gives the presentation

$$\Gamma_2 \cong \pi_1(S_2) \cong \langle a, b, c, d \mid [a, b][c, d] = 1 \rangle$$

where $[a, b] = aba^{-1}b^{-1}$. □

More generally, for any $g \geq 2$ choose a regular hyperbolic $4g$–gon with interior angles $\pi/(2g)$ and identify sides according to the cyclic boundary word $[a_1, b_1] \cdots [a_g, b_g]$. This yields a closed orientable surface S_g and an action of:

$$\pi_1(S_g) \cong \left\langle a_1, b_1, \ldots, a_g, b_g \;\middle|\; \prod_{i=1}^{g} [a_i, b_i] = 1 \right\rangle$$

freely, properly discontinuously, and cocompactly on $\mathbb{H}^2$.

Note that the corresponding argument for $g = 1$ requires a regular 4-gon with interior angles $\pi/2$, but that is the *Euclidean* angle measure of a regular 4-gon, so the construction needs to take a square P in $\mathbb{E}^2$ and realize its identification space as a torus with a curvature 0 metric, whose universal cover is $\mathbb{E}^2$.

2.6. Dehn's Algorithm for surface groups

We first state a theorem from geometry in its general form, using terms we have not defined:

THEOREM 2.6.1 (Gauss-Bonnet Theorem). *Let M be a compact Riemannian 2–manifold with piecewise smooth boundary with interior angles θ_i. Let K be the Gaussian curvature of M, and let k_g be the geodesic curvature of ∂M, and let $\chi(M)$ be the Euler characteristic of M. Then:*

$$\int_M K\, dA + \int_{\partial M} k_g\, ds + \sum_{i=1}^{n} \pi - \theta_i = 2\pi\chi(M)$$

When M has a metric defined to be locally isometric to $\mathbb{H}^2$, as in the previous section, $K \equiv -1$. When the boundary is piecewise geodesic the geodesic curvature of each smooth boundary segment is 0. If M is contractible then $\chi(M) = 1$. In particular, if M is a compact, simply connected region in $\mathbb{H}^2$ with piecewise geodesic boundary then the Gauss-Bonnet formula becomes:

$$\text{Area}(M) = (n-2)\pi - \sum_{i=1}^{n} \theta_i \tag{2}$$

For our regular geodesic hyperbolic octagon P with interior angles $\pi/4$, this gives:

$$\text{Area}(P) = 4\pi \tag{3}$$

DEFINITION 2.6.2. Let D be a finite, simply connected subcomplex of the octagon tiling of $\mathbb{H}^2$ that is not a single vertex or a single face, and let γ be the closed edge path obtained by traversing ∂D clockwise, starting from some vertex.

A *spur* is a backtracking pair of edges of γ. A *shell* is a face of D with more than half of its boundary as a consecutive arc in ∂D. Call that arc its *long boundary arc.*

LEMMA 2.6.3. *Let D be a disk in $\mathbb{H}^2$ tiled by finitely many copies of the regular geodesic hyperbolic octagon P with interior angles $\pi/4$. Either D is a single copy of P or it contains at least two shells.*

PROOF. Let F be the number of octagons in D. Let B be the number of edges on ∂D.

(2) and (3) give:

$$4\pi F = \text{Area}(D) = (B-2)\pi - \sum_{i=1}^{B} \theta_i$$

Since P has interior angles $\pi/4$, D has interior angles that are integer multiples of $\pi/4$, so each is at least $\pi/4$. Thus:

$$4\pi F \leq (B-2)\pi - B\pi/4$$

This gives:

$$16F \leq 3B - 8 \tag{4}$$

Suppose there is at most one face with more than half of its boundary on ∂D. That face contributes at most 8 edges to ∂D, and every other face contributes at most 4 edges to ∂D, so:

$$B \leq 8 + 4(F-1)$$

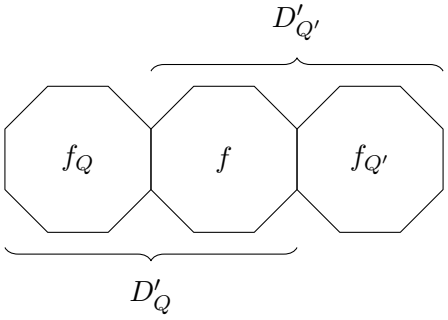
Combined with (4), this gives:

$$16F \leq 3(8 + 4(F-1)) - 8 = 12F + 4 \implies F \leq 1$$

Thus, either $F = 1$, in which case we are done, or there are at least 2 faces with more than half of their boundaries in ∂D.

We now improve the result to find at two shells by induction on the number of faces of D. The result is true if D has two faces, since then they share one edge and each face is a shell. Suppose now that we have established the lemma for all disks with strictly fewer faces than D. The previous argument gives at least two faces of D with more than half of their boundary on ∂D. Suppose f is such face that is not a shell. Then ∂f is subdivided into maximal arcs that are in ∂D and complementary segments in the interior of D, and there are at least two of each, since the boundary arcs combined contain at least 5 edges, but none of them individually contains that many edges. Thus, removing f disconnects D. For each component Q of $D \setminus f$,

consider $D'_Q := Q \cup f$. It is a disk tiled by copies of P with at least 2 faces, and strictly fewer faces than D. By the induction hypothesis, it has at least two shells. At most one of these is f. Let f_Q be the other. Since the various D'_Q intersect only in f, the long arc of ∂f_Q contained in $\partial D'_Q$ is also in ∂D. Since there are at least two components Q, this gives the desired two shells.



□

THEOREM 2.6.4 (Dehn's Algorithm). *Let $\Gamma_2 = \langle a, b, c, d \mid R \rangle$, with $R := [a, b][c, d]$. The following algorithm takes as input a word w in $a^{\pm 1}, b^{\pm 1}, c^{\pm 1}, d^{\pm 1}$ and in number of steps bounded above by $|w|$ decides whether or not $w =_{\Gamma_2} 1$.*

The algorithm is to repeatedly perform one of the following operations:

1. *Cancel a subword of the form xx^{-1} or $x^{-1}x$.*
2. *If w contains a subword u of length at least 5 that occurs in a cyclic permutation of R or R^{-1}, write that cyclic permutation as uv, and replace u by v^{-1}.*

When neither operation applies, the original word represents the identity if and only if the resulting word is empty.

PROOF. Both operations preserve the group element represented by w. Both operations strictly decrease the length of the current word. Thus the algorithm terminates.

It remains to show that if w is a nonempty word representing the identity, then one of the two operations applies.

Recall that the 1–skeleton of the octagon tiling of $\mathbb{H}^2$ is the Cayley graph of Γ_2 with respect to a, b, c, d. Since w represents the identity, it labels a nonconstant closed edge path γ in this graph.

Choose a shortest nonconstant closed subpath γ' of γ. If γ' is a spur, then operation (1) applies. Otherwise γ' is a simple closed edge path, so it bounds a disk D tiled by finitely many octagons. If D consists of a single octagon, then γ' contains at least five consecutive edges of its boundary, so operation (2) applies. Otherwise, Lemma 2.6.3 gives at least two shells of D, since its boundary is simple and therefore has no spurs. At most one of their long boundary arcs contains the initial vertex of γ' in its interior. Thus at least one shell gives a subpath of γ' consisting of at least five consecutive edges of an octagon, without wrapping around the initial vertex. The label of this subpath is a subword of w occurring in a cyclic permutation of R or R^{-1}. Hence operation (2) applies.

We have shown that every nonempty word representing the identity admits a shortening operation. Therefore the algorithm can terminate with a nonempty word only if that word represents a nonidentity element of Γ_2. □

2.7. Exercises

EXERCISE 2.1. Show that the Cayley transform

$$C(z) = i\frac{1+z}{1-z}$$

takes the Poincaré disk metric to $(dx^2 + dy^2)/y^2$ on the upper half-plane.

EXERCISE 2.2. Let p be a parabolic orientation-preserving isometry of $\mathbb{H}^2$, and let $R(z) = \bar{z}$. Show that $R \circ p$ is either a reflection in a geodesic or a glide reflection. Find examples where each possibility occurs.

EXERCISE 2.3. Check directly from the edge pairing of the octagon that all its vertices become a single vertex. Why does it matter that the total angle there is 2π?

EXERCISE 2.4. For the $4g$–gon construction, compute the Euler characteristic of the quotient from its numbers of vertices, edges, and faces.

EXERCISE 2.5. Why can no nonidentity element of the surface group Γ_2 act as an elliptic isometry?

EXERCISE 2.6 (Geodesics and tangent vectors). Let $p \in \mathbb{H}^2$ and let $0 \neq v \in T_p\mathbb{H}^2$. Show that, up to orientation-preserving reparametrization, there is a unique complete geodesic through p whose tangent vector at p is a positive multiple of v.

EXERCISE 2.7 (Angles of hyperbolic triangles). Recall the Gauss-Bonnet Theorem, Theorem 2.6.1, and its specialization (2).

(1) Use Gauss–Bonnet to prove that:
$$\text{Area}(T) = \pi - \alpha - \beta - \gamma$$
Deduce that:
$$\alpha + \beta + \gamma < \pi$$
(2) Conversely, suppose $\alpha, \beta, \gamma \in (0, \pi)$ satisfy $\alpha + \beta + \gamma < \pi$. Prove that there exists a hyperbolic geodesic triangle with these angles.
(3) Prove that this triangle is unique up to hyperbolic isometry.

EXERCISE 2.8 (Area of closed hyperbolic surfaces). Use the Gauss–Bonnet formula for a geodesic n–gon with interior angles $\alpha_1, \ldots, \alpha_n$:
$$\text{Area}(P) = (n-2)\pi - \sum_{j=1}^{n} \alpha_j$$
to compute the area of the regular hyperbolic $4g$–gon used in the construction of S_g. Deduce that:
$$\text{Area}(S_g) = 4\pi(g-1)$$

EXERCISE 2.9 (Ideal triangles and thin triangles). An *ideal triangle* in $\mathbb{H}^2$ is the closed convex region bounded by three bi-infinite hyperbolic geodesics joining three distinct points of $\partial\mathbb{H}^2$.

A geodesic triangle (finite or ideal) is called δ-*thin* if every point on each side lies at distance at most δ from the union of the other two sides.

(1) Show that any two ideal triangles are related by an isometry of $\mathbb{H}^2$.
(2) Let T be a finite geodesic triangle in the Poincaré disk. Show that some hyperbolic isometry takes one side of T into the real diameter and its opposite vertex onto the positive imaginary axis. Deduce that the image of T is contained in the ideal triangle Δ with vertices -1, 1, and i.
(3) Show that Δ is δ-thin for:
$$\delta = \text{arsinh}(1) = \log(1 + \sqrt{2})$$
Hint: Let g_+ and g_- be the sides of Δ joining i to 1 and -1, respectively. Show that for $x \in (-1, 1)$:
$$d_{\mathbb{H}}(x, g_+) = \text{arsinh}\left(\frac{1-x}{1+x}\right)$$
$$d_{\mathbb{H}}(x, g_-) = \text{arsinh}\left(\frac{1+x}{1-x}\right)$$

To compute these distances, use the disk isometry:

$$z \longmapsto \frac{z-x}{1-xz}$$

which sends x to 0. Use the symmetries of Δ to deal with the other two sides.

(4) Deduce that every finite geodesic triangle in $\mathbb{H}^2$ is δ-thin, with the same constant δ.

Hint: Given a point p on one side of a finite triangle, normalize the triangle as in part (b). Choose a point q on one of the other two sides of the enclosing ideal triangle with $d_{\mathbb{H}}(p,q) \leq \delta$. Explain why the geodesic segment $[p,q]$ must meet one of the other two sides of the finite triangle.

EXERCISE 2.10 (Area, circumference, and isoperimetry). Let B_R be the hyperbolic ball of radius R centered at 0 in the Poincaré disk. Recall that B_R is a Euclidean disk of radius $\tanh(R/2)$.

(1) Using the hyperbolic metric

$$ds = \frac{2|dz|}{1-|z|^2}$$

compute the hyperbolic circumference $L(R)$ of ∂B_R. Show that:

$$L(R) = 2\pi \sinh R$$

(2) Using the hyperbolic area element

$$dA = \frac{4\,dx\,dy}{(1-|z|^2)^2}$$

compute the hyperbolic area $A(R)$ of B_R. Show that:

$$A(R) = 2\pi(\cosh R - 1)$$

(3) Show that:

$$L(R)^2 = A(R)^2 + 4\pi A(R)$$

Compare this with the corresponding formula for Euclidean disks.

(4) Show that $A(R) \leq L(R)$ for all $R > 0$, and that:

$$\lim_{R\to\infty} \frac{L(R)}{A(R)} = 1$$

Explain how this differs from the behavior of Euclidean disks.

LECTURE 3

Basics of geometric group theory

3.1. Word metrics and Cayley graphs

In Lecture 1 we constructed an action of F_2 on the regular tree T_4, and in Lecture 2 we constructed an action of the surface group Γ_2 on $\mathbb{H}^2$ using a regular octagon. We will now see that these are instances of the same phenomenon: a group can be regarded as a metric space, and certain actions allow us to recover its geometry from that of the space on which it acts.

Let G be a group generated by a finite set S. We will assume $1 \notin S$ and $S \cap S^{-1}$ contains only those generators that have order 2. A *word in* S is an element of $(S \cup S^{-1})^*$, meaning a finite sequence $w = s_1^{\epsilon_1} \cdots s_n^{\epsilon_n}$ of elements of S or their inverses. Its length is $|w| = n$. The empty word has length 0 and represents the identity element of G.

Definition 3.1.1. The *word length* of $g \in G$ with respect to S is:

$$|g|_S := \min\{|w| \mid w =_G g,\, w \in (S \cup S^{-1})^*\}$$

The *word metric* associated to S is:

$$d_S(g,h) := |g^{-1}h|_S$$

Proposition 3.1.2. *The function d_S is a metric on G, and left multiplication by any element of G is an isometry of (G, d_S).*

Proof. Since the only word of length 0 is the empty word, $|g|_S = 0$ if and only if $g = 1$. Inverting a word and reversing its letters shows that $|g^{-1}|_S = |g|_S$. Concatenating shortest words for g and h shows that:

$$|gh|_S \leq |g|_S + |h|_S$$

These facts give positive definiteness, symmetry, and the triangle inequality. Finally:

$$d_S(kg, kh) = |(kg)^{-1}(kh)|_S = |g^{-1}h|_S = d_S(g,h)$$ □

Definition 3.1.3. The *Cayley graph* $\mathrm{Cay}(G,S)$ has vertex set G. For each $g \in G$ and $s \in S$ that does not have order 2, there is a directed edge from g to gs, labelled s. For each $g \in G$ and $s \in S$ that does have order 2, there is an undirected edge from g to gs, labelled s. We give each edge length 1 and use the resulting path metric on the underlying undirected graph.

Definition 3.1.4. To every edge path in $\mathrm{Cay}(G,S)$ we associate a *label* by concatenating the labels of the edges of the path according to the following conventions:

- An undirected edge labelled s contributes letter s.
- A directed edge labelled s traversed in the given direction contributes letter s.
- A directed edge labelled s traversed against its given direction contributes letter s^{-1}.

Lemma 3.1.5. *Given a vertex $g \in \mathrm{Cay}(G,S)$ and $w \in (S \cup S^{-1})^*$ there is a unique path in $\mathrm{Cay}(G,S)$ whose initial vertex is g and whose label is w. Its end vertex is the element of G represented by gw.*

Proof. Since $1 \notin S$ there is no loop at g. if $S' := \{s \in S \mid s^2 = 1\}$, then g has undirected incident edges labelled by the elements of S', outgoing directed edges labelled by the elements of $S \setminus S'$, and incoming directed edges labelled by the elements of $S \setminus S'$. Furthermore, we claim that all of these edges lead to different vertices. Edges leading to the same vertex would give $gs_1^{\epsilon_1} =_G gs_2^{\epsilon_2}$. Then $s_1 = s_2^{\epsilon_1\epsilon_2}$. Since $S \cap S^{-1} = S'$ the possibilities are $s_1 = s_2 \in S'$ and $\epsilon_1 = \epsilon_2 = 1$ or $s_1 = s_2 \in S \setminus S'$ and $\epsilon_1 = \epsilon_2$. In both cases, that means the two edges were not distinct.

Now argue by induction on $|w|$. If $|w| = 0$ then the path is the constant path at the vertex g, and its end vertex is $gw =_G g$. If the lemma is true for paths of length $\leq n$ and $|w| = n+1$ then write $w = w's^\epsilon$. There is a unique path from g to gw' labelled w', and there is a unique path starting at gw' labelled s^ϵ, by the previous paragraph. Their concatenation is the unique path starting at g and labelled by w, and it ends at gw. □

PROPOSITION 3.1.6.

(a) $\mathrm{Cay}(G,S)$ *is a connected, locally finite, geodesic metric graph.*
(b) The restriction of its metric to the vertices is d_S*.*
(c) Left multiplication defines an isometric action of G *on* $\mathrm{Cay}(G,S)$ *that is free and transitive on vertices.*

PROOF. Lemma 3.1.5 shows that $\mathrm{Cay}(G,S)$ has constant valence $|S'| + 2|S \setminus S'|$, and that edge paths correspond to elements of $(S \cup S^{-1})^*$ via their labels. For any $g \in G$, any word $w \in (S \cup S^{-1})^*$ representing g gives a path in $\mathrm{Cay}(G,S)$ starting at 1 and ending at g. The fact that the graph is a geodesic metric space follows from the construction giving all edges length 1.

That the restriction of the length metric to vertices is exactly d_S follows from Lemma 3.1.5.

Left multiplication by h takes an edge $g - gs$ to an edge $hg - hgs$, preserving its label and direction, so it gives an automorphism of the labelled, directed graph structure, which is stronger than being an isometry.

The action on vertices is free because $gh = h$ if and only if $g = 1$, and it is transitive because $(hg^{-1})g = h$. □

REMARK 3.1.1. As we have defined the Cayley graph, the action of G on $\mathrm{Cay}(G,S)$ is not free on the entire graph if S contains order 2 elements, although it is free on the vertices: an order 2 element $s \in S$ inverts the edge $1-s$, fixing its midpoint. A common alternative definition is to define a directed edge labelled s from 1 to s and another directed edge labelled s from s to 1. Then the s action exchanges these two edges. This makes the action of G on $\mathrm{Cay}(G,S)$ free, at the expense of introducing bigons, so complicating the topology of the underlying graph.

EXAMPLE 3.1.7 (Free groups and surface groups). By the construction in Lecture 1, the Cayley graph of F_2 with its standard generators is the tree T_4:

$$\mathrm{Cay}(F_2, \{a,b\}) \cong T_4$$

Reduced words label geodesic paths starting at the identity.

In Lecture 2, the 1–skeleton of the octagon tiling of $\mathbb{H}^2$ was identified with the Cayley graph

$$\mathrm{Cay}(\Gamma_2, \{a,b,c,d\})$$

where

$$\Gamma_2 = \langle a,b,c,d \mid [a,b][c,d] = 1 \rangle$$

A word representing the identity labels a closed edge path based at 1. Thus relations in a group produce closed paths in its Cayley graph; for example, the surface relator labels the boundary of an octagon.

3.2. Changing generating sets

Recall that a map $f\colon X \to Y$ is *L–Lipschitz* if $d_Y(f(x), f(x')) \leq L d_X(x,x')$ for all $x, x' \in X$. A bijection is a *L–biLipschitz equivalence* if both it and its inverse are L–Lipschitz. It is a *biLipschitz equivalence* if there exists an L such that it is an L–biLipschitz equivalence.

PROPOSITION 3.2.1. *Let S and T be finite generating sets of G. The identity map on G is a biLipschitz equivalence between (G, d_S) and (G, d_T). Specifically, there exists $L \geq 1$ such that for all $g, h \in G$:*

$$L^{-1} d_S(g,h) \leq d_T(g,h) \leq L d_S(g,h)$$

PROOF. Let:

$$L := \max\{\max\{|s|_T \mid s \in S\}, \max\{|t|_S \mid t \in T\}\}$$

If $w \in (S \cup S^{-1})^*$ of length n then each letter of w can be rewritten as a word of length at most L in $(T \cup T^{-1})^*$, and their concatenation is a word w' of length at most nL such that $w =_G w'$. Thus, $d_T(g,h) \leq Ld_S(g,h)$. The analogous argument gives $d_S(g,h) \leq Ld_T(g,h)$. □

3.3. Quasi-isometries

A subset $Z \subset Y$ is *A–dense* if every point of Y is at distance at most A from some point of Z. The subset Z is *coarsely dense* if it is A–dense for some A.

DEFINITION 3.3.1. Let (X, d_X) and (Y, d_Y) be metric spaces. A map $\phi\colon X \to Y$ is a *(L,A)–quasiisometric embedding* if for all $x, x' \in X$:

$$\frac{1}{L} d_X(x,x') - A \leq d_Y(\phi(x), \phi(x')) \leq Ld_X(x,x') + A$$

The map is an *(L,A)–quasiisometry* if it is an (L,A)–quasiisometric embedding and, in addition, $\phi(X)$ is A–coarsely dense in Y.

EXAMPLE 3.3.2. An L–biLipschitz equivalence is an $(L,0)$–quasiisometry, so Proposition 3.2.1 shows that for any finite generating sets S and T of a group G, the identity map on G is a quasiisometry between (G, d_S) and (G, d_T).

EXAMPLE 3.3.3. The inclusion $\mathbb{Z} \hookrightarrow \mathbb{R}$ is an isometric embedding with $\frac{1}{2}$–dense image, hence is a quasi-isometry. The nearest-integer map $\mathbb{R} \to \mathbb{Z}$ is a quasi-isometry in the opposite direction, although it is not continuous.

EXAMPLE 3.3.4. The inclusion $\mathbb{Z}^2 \hookrightarrow \mathbb{R}^2$ is a quasi-isometry, where $\mathbb{Z}^2$ has its standard word metric and $\mathbb{R}^2$ its Euclidean metric. Indeed, for $u \in \mathbb{R}^2$:

$$\|u\|_2 \leq \|u\|_1 \leq \sqrt{2}\,\|u\|_2.$$

Moreover, every point of $\mathbb{R}^2$ lies at Euclidean distance at most $\sqrt{2}/2$ from a lattice point.

EXAMPLE 3.3.5 (Quasi-isometric embedding versus quasi-isometry). The inclusion $\mathbb{Z} \to \mathbb{R}^2$ given by $n \mapsto (n,0)$ is a quasi-isometric embedding. It is not a quasi-isometry, since the points $(0,t)$ lie arbitrarily far from its image.

REMARK 3.3.1. Every bounded metric space is quasi-isometric to a point. The additive constant A in the definition is essential for this observation, and more generally allows us to ignore geometry at bounded scales.

As a corollary, every finite group is quasiisometric to the trivial group.

EXAMPLE 3.3.6. If H is a finitely generated, finite index subgroup of a finitely generated group G then for any finite generating sets S of G and T of H, the inclusion of H into G is a quasiisometry from (H, d_T) into (G, d_s).

Write $G = Hg_0 \sqcup \cdots \sqcup Hg_n$, with $g_0 = 1$. Then $R := \{g_1, \ldots, g_n\} \cup T$ is a finite generating set for G, and by Example 3.3.2 $(G,R) \simeq_{\mathrm{QI}} (G,S)$. A composition of quasiisometries is a quasiisometry, so it suffices to show that $\iota\colon H \hookrightarrow G$ gives a quasiisometry $(H,T) \to (G,R)$.

The map ι has 1–dense image, since for every $g \in G$ there exist $h_g \in H$ and i such that $g = h_g g_i$, and $g_i \in R$ unless $i = 0$, in which case $g = h_g$ is already in H.

The map is distance non-increasing, since $T \subset R$. The only worry is that it may shrink distances: there might be a long word in $(T \cup T^{-1})^*$ that represents the same element of H as a much shorter word in $(R \cup R^{-1})^*$. We argue that that cannot happen.

Define:

$$L := \max\{|w|_T \mid w \text{ is a word of length at most 3 in } (R \cup R^{-1})^* \text{ such that } w \in H\}$$

Suppose $h \in H$. Take a shortest word $w \in (R \cup R^{-1})^*$ representing h. If $w = r_1^{\epsilon_1} \cdots r_m^{\epsilon_m}$ then let x_0 be the vertex 1 in $\mathrm{Cay}(G,R)$ and let x_i be the vertex at the end of the path starting at x_0 labelled by $r_1^{\epsilon_1} \cdots r_i^{\epsilon_i}$, so $h = x_m$. Let y_i be a vertex of $\mathrm{Cay}(G,R)$ corresponding to an element

of H at minimal distance from x_i. We have already shown that $d_R(x_i, y_i) \in \{0, 1\}$. Since $d_R(x_i, x_{i+1}) = 1$, this gives $d_R(y_i, y_{i+1}) \leq 3$, so, by definition of L, we have $d_T(y_i, y_{i+1}) \leq L$. Thus, $d_T(1, h) \leq mL$. Conclude:

$$d_R(1, h) \leq d_T(1, h) \leq L d_R(1, h)$$

3.4. Geometric group actions

A metric space is called *proper* if its closed balls are compact.

DEFINITION 3.4.1. Let G act by isometries on a proper geodesic metric space X.
The action is *cocompact* if there exists a compact $K \subset X$ such that $GK := \cup_{g \in G} gK = X$.
The action is *properly discontinuous* if for every compact subset K of X:

$$|\{g \in G \mid gK \cap K \neq \emptyset\}| < \infty$$

The action is *geometric* if it is properly discontinuous and cocompact.

THEOREM 3.4.2 (Fundamental Theorem of Geometric Group Theory/Švarc–Milnor Lemma). *If G acts geometrically on a proper geodesic metric space X then G is finitely generated and for any finite generating set S of G the orbit map $g \mapsto g.o$ is a quasiisometry $(G, d_S) \to (X, d_X)$, for any basepoint $o \in X$.*

PROOF. Fix $o \in X$ and a compact set C such that $GC = X$, which exists by cocompactness of the action. There is some translate gC of C such that $o \in gC$, but $GC = G(gC)$, so we might as well assume that $o \in C$. Let R be large enough that $\bar{B}_R(o)$ contains C. Define:

$$S := \{g \in G \mid d_X(o, go) \leq 3R\} = \{g \in G \mid \bar{B}_{3R/2}(o) \cap g\bar{B}_{3R/2}(o) \neq \emptyset\}$$

Since X is proper, the closed balls in the definition are compact, so proper discontinuity of the action makes S finite.

For any $h \in G$ let $\gamma\colon [0, d_X(o, ho)] \to X$ be a (unit speed) geodesic from o to ho. Set $t_0 := 0$ and $g_0 := 1$. Let t_1 be the last time such that $d_X(\gamma(t_1), g_0 o) = 2R$, or set $t_1 := d_X(o, ho)$ if no such time exists. There exists a $g_1 \in G$ such that $\gamma(t_1) \in g_1 C \subset g_1 \bar{B}_R(o) = \bar{B}_R(g_1 o)$. Continue inductively: given t_i and g_i let t_{i+1} be the last time in $(t_i, d_X(o, ho)]$ such that $d_X(\gamma(t_{i+1}), g_i o) = 2R$, or set $t_{i+1} := d_X(o, ho)$ if no such time exists. Then take g_{i+1} such that $\gamma(t_{i+1}) \in \bar{B}_R(g_{i+1} o)$, choosing $g_{i+1} := h$ if $t_{i+1} = d_X(o, ho)$.

This process stops in $n + 1 \leq \frac{d_X(o,ho)}{R} + 1$ steps, since if $t_{i+1} \neq d_X(o, ho)$ then:

$$d_X(\gamma(t_i), \gamma(t_{i+1})) \geq d_X(\gamma(t_{i+1}), g_i o) - d_X(g_i o, \gamma(t_i)) \geq 2R - R = R$$

Now:

$$d_X(g_i^{-1} g_{i+1} o, o) = d_X(g_i o, g_{i+1} o) \leq 3R \implies g_i^{-1} g_{i+1} \in S$$

But:

$$h = g_{n+1} = g_0^{-1} g_1 \cdot g_1^{-1} g_2 \cdots g_n^{-1} g_{n+1}$$

This expresses h as a product of elements of S, which shows S is a finite generating set of G. Moreover, $|h|_S \leq n + 1 \leq \frac{d_X(o,ho)}{R} + 1$.

Conversely, suppose $s_1 \cdots s_m$ is a shortest word in S^* representing h. For all i, group invariance and the definition of S give:

$$d_X(s_1 \cdots s_i o, s_1 \cdots s_{i+1} o) = d_X(o, s_{i+1} o) \leq 3R$$

Thus:

$$d_X(o, ho) \leq \sum_{i=1}^{m} d_X(o, s_i o) \leq 3Rm \implies d_X(o, ho) \leq 3R|h|_S$$

Conclude:

$$Rd_S(g, h) - R = R|g^{-1}h|_S - R \leq d_X(o, g^{-1}ho) = d_X(go, ho) \leq 3R|g^{-1}h|_S = 3Rd_S(g, h) \quad \square$$

3.5. Exercises

All metric spaces in these exercises are assumed to be nonempty. The metrics on finitely generated groups are word metrics associated to finite generating sets, unless another metric is specified.

Two maps $\phi, \phi' \colon X \to Y$ are *at finite distance* if

$$\sup_{x \in X} d_Y(\phi(x), \phi'(x)) < \infty.$$

Write $\phi \sim \phi'$ in this case. A *quasiinverse* of a quasiisometry $\phi \colon X \to Y$ is a quasiisometry $\psi \colon Y \to X$ such that $\psi \circ \phi \sim \mathrm{id}_X$ and $\phi \circ \psi \sim \mathrm{id}_Y$.

Exercise 3.1 (Constructing a quasiinverse). Let $\phi \colon X \to Y$ be an (L, A)–quasiisometry, where $L \geq 1$ and $A \geq 0$.

(a) Construct a map $\psi \colon Y \to X$ satisfying $d_Y(\phi(\psi(y)), y) \leq A$ for every $y \in Y$.
(b) Show that $\psi \circ \phi \sim \mathrm{id}_X$, and prove that ψ is a quasiisometry. Give constants in terms of L and A.
(c) Prove that any two quasiinverses of ϕ are at finite distance.

Exercise 3.2 (Quasiisometry is an equivalence relation).

(a) Suppose $\phi \colon X \to Y$ is a quasiisometry and $\phi' \colon X \to Y$ is at finite distance from ϕ. Prove that ϕ' is also a quasiisometry.
(b) Let $\phi \colon X \to Y$ and $\psi \colon Y \to Z$ be quasiisometries with constants (L_1, A_1) and (L_2, A_2), respectively. Prove that $\psi \circ \phi$ is a quasiisometry, giving constants in terms of L_1, A_1, L_2, A_2.
(c) Define $X \simeq_{\mathrm{QI}} Y$ to mean that there exists a quasiisometry $X \to Y$. Prove that this relation is reflexive, symmetric, and transitive.

Exercise 3.3 (The quasiisometry group). Let $\mathrm{QI}(X)$ be the set of equivalence classes of quasiisometries $X \to X$, where two maps are equivalent if they are at finite distance.

(a) Verify that finite distance is an equivalence relation. Prove that

$$[\phi]\,[\psi] := [\phi \circ \psi]$$

is well-defined and makes $\mathrm{QI}(X)$ into a group. Identify its identity element and inverses.
(b) Let $\phi_0 \colon X \to Y$ be a quasiisometry with quasiinverse ψ_0. Show that

$$[\phi] \longmapsto [\phi_0 \circ \phi \circ \psi_0]$$

defines an isomorphism $\mathrm{QI}(X) \to \mathrm{QI}(Y)$, independent of the choice of ψ_0.
(c) For $a \in \mathbb{R} \setminus \{0\}$ and $b \in \mathbb{R}$, let $\phi_{a,b}(x) = ax + b$. Determine exactly when $[\phi_{a,b}] = [\phi_{a',b'}]$ in $\mathrm{QI}(\mathbb{R})$. Identify the subgroup represented by these affine maps.

Exercise 3.4 (Coarse connectivity and finite generation). A metric space X is *coarsely connected* if there exists $R > 0$ such that any two points can be joined by a finite sequence $x_0, \ldots, x_n$ in X satisfying $d_X(x_{i-1}, x_i) \leq R$ for every i.

(a) Show that geodesic metric spaces are coarsely connected, and that coarse connectivity is a quasiisometry invariant. Give an example of a coarsely connected space that is not connected.
(b) Let G be a group equipped with a left-invariant metric d such that every ball of finite radius contains finitely many elements. Prove that (G, d) is coarsely connected if and only if G is finitely generated. Deduce that, for groups equipped with such metrics, finite generation is a quasiisometry invariant.
(c) *Optional: constructing the metric.* Suppose G is countable, and choose a sequence $g_1, g_2, \ldots$ containing every element of G. Assign weight n to the letters g_n and g_n^{-1}, and put

$$\ell(g) := \min \left\{ \sum_{j=1}^{k} n_j \;\middle|\; g = g_{n_1}^{\epsilon_1} \cdots g_{n_k}^{\epsilon_k}, \quad \epsilon_j \in \{-1, 1\} \right\}.$$

The empty product is allowed and has weight 0. Prove that $d(g,h) := \ell(g^{-1}h)$ is a left-invariant metric with finite balls.

EXERCISE 3.5 (Homomorphisms that are quasiisometries).

(a) Let N be a finite normal subgroup of a finitely generated group G. Prove that the quotient homomorphism $q\colon G \to G/N$ is a quasiisometry.
(b) More generally, let $\phi\colon G \to H$ be a homomorphism between finitely generated groups. Prove that ϕ is a quasiisometry if and only if its kernel is finite and its image has finite index in H.
(c) Let D_∞ be the subgroup of $\mathrm{Isom}(\mathbb{R})$ generated by $r(x) = x+1$ and $s(x) = -x$. Show that $\mathbb{Z}$, $\mathbb{Z} \times (\mathbb{Z}/2\mathbb{Z})$, and D_∞ are quasiisometric. Deduce that being Abelian and being torsion-free are not quasiisometry invariants.

EXERCISE 3.6 (Growth as a quasiisometry invariant). Let G and H be finitely generated groups. For $r \geq 0$, put:

$$\beta_G(r) := |\overline{B}_r(1_G)| \qquad \beta_H(r) := |\overline{B}_r(1_H)|$$

(a) If $\phi\colon G \to H$ is a quasiisometric embedding, show that the cardinalities of its fibers $\phi^{-1}(h)$ have a uniform finite upper bound.
(b) If G and H are quasiisometric, prove that there is a constant $C \geq 1$ such that, for all $r \geq 0$:
$$\beta_G(r) \leq C\beta_H(Cr + C) \qquad \beta_H(r) \leq C\beta_G(Cr + C)$$
(c) For a fixed $d \geq 0$, prove that the property
$$\beta_G(r) \leq K(r+1)^d \qquad \text{for all } r \geq 0$$
for some constant K is a quasiisometry invariant. Show also that having two-sided bounds
$$k(r+1)^d \leq \beta_G(r) \leq K(r+1)^d \qquad \text{for all } r \geq 0$$
for some $k, K > 0$ is a quasiisometry invariant.
(d) Use the standard generators to estimate the growth of $\mathbb{Z}^m$ and to count the elements in a ball of integer radius in F_2. Deduce that $\mathbb{Z}^m$ and $\mathbb{Z}^n$ are not quasiisometric when $m \neq n$, and that F_2 is not quasiisometric to any $\mathbb{Z}^n$.

EXERCISE 3.7 (From a group to its quasiisometry group). Let G be a finitely generated group. For $g \in G$, define $L_g(x) := gx$ and $R_g(x) := xg$.

(a) Show that L_g is an isometry and that R_g is a quasiisometry at finite distance from the identity. Deduce that L_g and the inner automorphism $x \mapsto gxg^{-1}$ represent the same element of $\mathrm{QI}(G)$.
(b) Prove that $g \mapsto [L_g]$ is a homomorphism $G \to \mathrm{QI}(G)$ whose kernel consists exactly of those elements with finite conjugacy class.
(c) Determine this kernel when G is Abelian and when $G = F_2$. *Hint for F_2:* use reduced words or axes in its Cayley tree.

EXERCISE 3.8 (Two unbounded complementary components). Let S be any finite generating set of $\mathbb{Z}$, and equip $\mathrm{Cay}(\mathbb{Z}, S)$ with its path metric, giving every edge length 1. Prove that there exists $R_0 \geq 0$ such that, for every $R \geq R_0$, the complement

$$\mathrm{Cay}(\mathbb{Z}, S) \setminus \overline{B}_R(0)$$

has exactly two unbounded connected components. Hint: fix a finite path from 0 to 1 and consider the $\mathbb{Z}$ translates of it.

EXERCISE 3.9 (Taming quasigeodesics). Let X be a geodesic metric space. Show that given L and A there exist L', A' and H such that for every (L, A)–quasigeodesic segment $\gamma\colon [0,T] \to X$ there is a continuous (L', A')–quasigeodesic $\gamma'\colon [0,T] \to X$ with the same endpoints as γ and at Hausdorff distance at most H from γ. *Hint:* Consider $[0,T] \cap \mathbb{Z}$ and connect the dots.

Part 2

Hyperbolic groups

LECTURE 4

Coarse geometry of hyperbolic spaces

4.1. Thin triangles and the Gromov product

DEFINITION 4.1.1. Let X be a geodesic metric space and $\delta \geq 0$. A *geodesic triangle* consists of geodesic segments γ_0, γ_1, and γ_2 such that the terminal endpoint of γ_i is the initial endpoints of γ_{i+1} for each $i \in \{0, 1, 2\}$, with subscripts mod 3. The triangle is *δ–thin* if for each i the side γ_i is contained in $\bar{N}_\delta(\gamma_{i+1} \cup \gamma_{i+1})$. The space X is *δ–hyperbolic* if every geodesic triangle in X is δ–thin. It is *hyperbolic* if there exists δ such that it is δ–hyperbolic.

This notion of coarse hyperbolicity is sometimes called "Gromov hyperbolicity" to distinguish it from the smooth hyperbolicity of spaces like $\mathbb{H}^n$.

EXAMPLE 4.1.2.

(a) Every tree is 0–hyperbolic by Theorem 1.3.8.
(b) Every bounded geodesic space is hyperbolic: take δ to be the diameter of the space.
(c) The hyperbolic plane $\mathbb{H}^2$ is δ–hyperbolic for $\delta = \log(1 + \sqrt{2})$, by Exercise 2.9.
(d) The Euclidean plane is not hyperbolic. For each $R > 0$ take a triangle with vertices $(0, 0)$, $(2R, 0)$, and $(0, 2R)$. The midpoint (R, R) of the side opposite $(0, 0)$ has distance R from the union of the other two sides.

LEMMA 4.1.3 (Geodesic bigons). *If X is δ–hyperbolic, any two geodesic segments with the same endpoints have Hausdorff distance at most δ.*

PROOF. Let α and $\bar{\beta}$ be geodesics from x to y. Let γ be the constant geodesic at x. Then α, β, and γ form a geodesic triangle. We have $\alpha \subset \bar{N}_\delta(\beta \cup \gamma) = \bar{N}_\delta(\beta)$, and similarly $\beta \subset \bar{N}_\delta(\alpha)$. □

There is a converse to Lemma 4.1.3 for graphs. It is much deeper:

THEOREM 4.1.4 (Papasoglu [**1**]). *Let Γ be a connected graph with the path metric obtained by assigning length 1 to every edge. If there exists $\delta \geq 0$ such that any two geodesic segments in Γ with the same endpoints have Hausdorff distance at most δ, then Γ is hyperbolic.*

The Euclidean plane shows that Theorem 4.1.4 cannot be true for arbitrary geodesic metric spaces.

Recall from Definition 1.4.1 that the Gromov product of $x, y \in X$ based at $o \in X$ is:

$$(x \mid y)_o = \frac{1}{2}(d(o, x) + d(o, y) - d(x, y))$$

LEMMA 4.1.5 (Fellow traveling). *Suppose X is δ–hyperbolic. Let $\alpha\colon [0, A] \to X$ and $\beta\colon [0, B] \to X$ be geodesics with $\alpha(0) = \beta(0) = o$. For all $0 \leq t \leq (x \mid y)_o$ we have:*

$$d(\alpha(t), \beta(t)) \leq 4\delta$$

PROOF. Let $x := \alpha(A)$ and $y := \beta(B)$, and let $\gamma\colon [0, C] \to X$ be a geodesic from x to y. Let $D := (x \mid y)_o$. It follows from the triangle inequality that $D \leq \min\{A, B\}$, so for $0 \leq t \leq D$ the points $\alpha(t)$ and $\beta(t)$ are defined.

If $D \leq \delta$ then for all $0 \leq t \leq D$ we have $d(\alpha(t), \beta(t)) \leq 2\delta$, and we are done. Suppose not.

Let z be an arbitrary point on γ. The triangle inequality gives:

$$d(o, x) \leq d(o, z) + d(z, x)$$
$$d(o, y) \leq d(o, z) + d(z, y)$$

Adding, and using that fact that z sits on a geodesic between x and y, gives:

$$A + B \leq 2d(o, z) + C$$

Thus:

$$D = \frac{1}{2}(A + B - C) \leq d(o, z)$$

This implies that for $0 \leq s < D - \delta$ we have $d(\alpha(s), \gamma) > \delta$, so there exists s' such that $d(\alpha(s), \beta(s')) \leq \delta$. By the triangle inequality, $|s - s'| \leq \delta$, so $d(\alpha(s), \beta(s)) \leq 2\delta$. By continuity, $d(\alpha(D - \delta), \beta(D - \delta)) \leq 2\delta$. For $t \in (D - \delta, D]$ we then have:

$$d(\alpha(t), \beta(t)) \leq d(\alpha(D - \delta), \beta(D - \delta)) + 2\delta \leq 4\delta \qquad \square$$

The next result generalizes Exercise 1.4 from the tree case.

PROPOSITION 4.1.6 (4-point Gromov product inequality). *Suppose X is δ–hyperbolic. For all $o, x, y, z \in X$:*

$$(x \mid z)_o \geq \min\{(x \mid y)_o, (y \mid z)_o\} - 4\delta$$

PROOF. Set $D := \min\{(x \mid y)_o, (y \mid z)_o\}$ and choose geodesics α, β, and γ from o to x, y, and z, respectively. By Lemma 4.1.5, for $0 \leq t \leq D$ we have $d(\alpha(t), \beta(t)) \leq 4\delta$ and $d(\beta(t), \gamma(t)) \leq 4\delta$, so $d(\alpha(t), \gamma(t)) \leq 8\delta$. In particular, this is true at $t = D$. Then:

$$\begin{aligned} d(x, z) &\leq d(x, \alpha(D)) + d(\alpha(D), \gamma(D)) + d(\gamma(D), z) \\ &\leq d(o, x) - D + 8\delta + d(o, z) - D \end{aligned}$$

Rearranging gives:

$$D - 4\delta \leq \frac{1}{2}(d(o, x) + d(o, z) - d(x, z)) = (x \mid z)_o \qquad \square$$

4.2. The Morse lemma

We extend thinness from triangles to arbitrary n-gons:

LEMMA 4.2.1 (Logarithmic thinness). *Suppose X is δ–hyperbolic. For $n \geq 2$, every side of a geodesic n–gon is contained in the closed $(1 + \lceil \log_2(n-1) \rceil)\delta$–neighborhood of the union of its other sides.*

PROOF. We argue by induction on n. For $n = 2$ the claim is Lemma 4.1.3. Suppose that $n > 2$ and the estimate is true for all geodesic polygons with fewer than n sides. Let α_0, $\alpha_1, \ldots, \alpha_{n-1}$ be the sides of a geodesic n-gon. Let x and y be the terminal and initial endpoints of α_0. Then $\alpha_1 + \alpha_2 + \cdots + \alpha_{n-1}$ is a path from x to y. Let $k := \lceil (n-1)/2 \rceil$. Then $1 \leq k \leq n$. Let z be the terminal endpoint of α_k. Let β be a geodesic from z to x, and let β' be a geodesic from y to z. Then $\beta, \alpha_1, \ldots, \alpha_k$ is a geodesic $(k+1)$-gon, $\beta', \alpha_{k+1}, \ldots, \alpha_{n-1}$ is a geodesic $(n-k)$–gon, and $\bar{\alpha}_0 + \beta' + \beta$ is a geodesic triangle. Every point a of α_0 is within δ of a point a' of either β or β'. Applying the induction hypothesis and the fact that $\lceil \frac{n-1}{2} \rceil = k = \max\{k+1, n-k\} - 1$, we have that a' is within $\delta + (1 + \lceil \log_2 \lceil \frac{n-1}{2} \rceil \rceil)\delta$ of $\bigcup_{i=1}^{n-1} \alpha_i$. Thus, it remains to show:

$$1 + \lceil \log_2 \lceil \frac{n-1}{2} \rceil \rceil = \lceil \log_2 2 \lceil \frac{n-1}{2} \rceil \rceil \leq \lceil \log_2 n - 1 \rceil$$

In fact, for integral $n > 2$ this is an equality. This can be proven as follows. First observe $\lceil \log_2 n - 1 \rceil = m$ for $2^{m-1} + 1 < n \leq 2^m + 1$. Since $n \geq 3$ is an integer:

$$\begin{aligned} \lceil \log_2 n - 1 \rceil = m &\iff 2^{m-1} < n - 1 \leq 2^m \\ &\iff 2^{m-1} + 1 < n \leq 2^m + 1 \\ &\iff \left(n \text{ even}, 2^{m-1} + 2 \leq n \leq 2^m\right) \vee \left(n \text{ odd}, 2^{m-1} + 3 \leq n \leq 2^m + 1\right) \end{aligned}$$

Now check that $\lceil \log_2 2 \lceil \frac{n-1}{2} \rceil \rceil = m$ has the exact same two cases. $\square$

COROLLARY 4.2.2 (Logarithmic detours). *Suppose X is δ–hyperbolic, γ is a rectifiable path from x to y, and α is a geodesic from x to y. Put $m = \max\{1, \lceil \text{length}(\gamma) \rceil\}$. Then every point p of α satisfies:*

$$d(p, \gamma) \leq 1 + \delta(1 + \lceil \log_2 m \rceil)$$

PROOF. Subdivide γ into at most m subpaths of length at most 1. Replace each subpath by a geodesic segment joining its endpoints. These segments, together with α, form a geodesic polygon with at most $m+1$ sides. By Lemma 4.2.1, every point of α lies at distance at most $\delta(1 + \lceil \log_2 m \rceil)$ from one of the new sides. Each new side has length at most 1, so it lies in the 1–neighborhood of the original path c. □

Recall from Lemma 1.3.6 that it is impossible for a path γ to travel in a tree and avoid a point p on the geodesic α between the start and endpoints. Corollary 4.2.2 says that in a hyperbolic space it is possible, but expensive: if γ avoids the ball of radius r about $p \in \alpha$ then the length of γ is bounded below by an exponential function of r.

THEOREM 4.2.3 (Morse lemma/stability of quasigeodesics). *For all δ, λ, and ϵ there exists μ such that if X is a δ–hyperbolic geodesic space, $\gamma\colon [0, T] \to X$ is a (λ, ϵ)–quasigeodesic, and α is a geodesic with the same endpoints as γ, then:*

$$d_{Haus}(\gamma, \alpha) \leq \mu$$

PROOF. First we fix up γ. If $T = 0$ there is nothing to prove. If not, we may assume T is a positive integer by extending γ to be constant on $[T, \lceil T \rceil]$, at the cost of replacing ϵ by $\max\{\epsilon, 1\}$. Then for each $s \in [0, T) \cap \mathbb{Z}$ replace the subsegment of γ on $[s, s+1]$ by a constant speed geodesic path $[s, s+1] \to X$ from $\gamma(s)$ to $\gamma(s+1)$. Since $d(\gamma(s), \gamma(s+1)) \leq \lambda + \epsilon$, it follows that for all $s, t \in [0, T]$ the length of the new path over the interval $[s, t]$ is at most $(\lambda+\epsilon)|t-s|$. As in Exercise 3.9, the new path is a quasigeodesic whose constants depend only on λ and ϵ, it has the same endpoints as γ, and it is Hausdorff equivalent to γ. Thus, but adjusting the constants accordingly, we may assume without loss of generality that $\gamma\colon [0, T] \to X$ is a piecewise geodesic (λ, ϵ–quasigeodesic path and there is a K depending on λ and ϵ such that for all $s, t \in [0, T]$ we have that the length of γ over the interval $[s, t]$ is at most $K|t - s|$.

Let $D := \max_{p \in \alpha} d(p, \gamma)$. Assume $\alpha\colon [0, d(\gamma(0), \gamma(T))] \to X$ is parameterized with unit speed. Choose t such that $d(\alpha(t), \gamma) = D$. Let $y := \alpha(\max\{0, t - 2D\})$ and $z := \alpha(\min\{t + 2D, |\alpha|\})$. Choose $s_y, s_z \in [0, T]$ such that for $y' := \gamma(s_y)$ and $z' := \gamma(s_z)$ we have $d(y, y') \leq D$ and $d(z, z') \leq D$. Observe that no geodesic from y to y' enters the D–ball about $\alpha(t)$. Likewise, no geodesic from z to z' enters that ball either. Now, $d(\gamma(s_y), \gamma(s_z)) \leq 6D$, so the quasigeodesic bounds imply $|s_y - s_z| \leq \lambda(6D + \epsilon)$, so the length of γ from y' to z' is at most $K\lambda(6D + \epsilon)$, and the choice of t guarantees that this subsegment of γ does not enter the D–ball about $\alpha(t)$. The concatenation of a geodesic from y to y', the subpath of γ, and a geodesic from z' to z is a path of length at most $2D + K\lambda(6D + \epsilon)$ that does not enter the D–ball about $\alpha(t)$. Corollary 4.2.2 then gives:

$$D \leq 1 + \delta(1 + \lceil \log_2 \lceil 2D + K\lambda(6D + \epsilon) \rceil \rceil)$$

The left-hand side grows linearly and the right-hand side growth logarithmically, so there is some uniform upper bound $D \leq D_0 = D_0(\delta, \lambda, \epsilon, K)$. This gives one direction of the desired Hausdorff distance bound.

In the other direction, let $(a, b) \subset [0, T]$ be a maximal interval on which for all $s \in (a, b)$ we have $d(\gamma(s), \alpha) > D$. If no such intervals exist we are done: take $\mu := D_0$. Assume not, so assume we have such an interval (a, b). Every point $c \in [a, b]$ is within $(b-a)/2$ of either a or b, so $\min\{d(\gamma(c), \gamma(a)), d(\gamma(c), \gamma(b))\} \leq \lambda(b-a)/2 + \epsilon$. Since $d(\gamma(a), \alpha) = d(\gamma(b), \alpha) = D$, we get:

$$d(\gamma(c), \alpha) \leq \lambda(b-a)/2 + \epsilon + 2D \tag{5}$$

Thus, we are done if we can bound $|b - a|$ uniformly.

Let s be the last time in $[0, |\alpha|]$ such that there exists $t_0 \in [0, a]$ with $d(\alpha(s), \gamma(t_0)) \leq D$. First suppose $s < |\alpha|$. For every $s' \in (s, |\alpha|]$ we have $d(\alpha(s'), \gamma([0, a])) > D$. By our choice of (a, b) we also have $d(\alpha(s'), \gamma((a, b))) > D$. However, $d(\alpha(s'), \gamma) \leq D$, so $\alpha(s')$ is D–close to

some point of $\gamma([b, |\alpha|])$. Since this is true for every $s' > s$, conclude there exists $t_1 \in [b, T]$ such that $d(\alpha(s), \gamma(t_1)) \leq D$. Then $d(\gamma(t_0), \gamma(t_1)) \leq 2D$, so:

$$|b - a| \leq |t_1 - t_0| \leq \lambda(2D + \epsilon) \tag{6}$$

If $s = |\alpha|$ then $\alpha(s) = \gamma(T)$. We have $d(\gamma(t_0), \gamma(T)) \leq D$, so $|b - a| \leq |T - t_0| \leq \lambda(D + \epsilon)$. This bound on $|b - a|$ is smaller than in the previous case, so we ignore it. Combine (5) and (6) to get:

$$d(\gamma(c), \alpha) \leq \lambda^2(2D + \epsilon/2) + 2D + \epsilon < (\lambda^2 + 1)(2D + \epsilon)$$

Therefore, (after having enlarged λ and ϵ in the fix-up step) it suffices to take:

$$\mu := (\lambda^2 + 1)(2D_0 + \epsilon)$$ □

4.3. Quasiisometry invariance of hyperbolicity

THEOREM 4.3.1. *Suppose $\phi\colon X \to Y$ is a (λ, ϵ)–quasiisometric embedding between geodesic metric spaces. Suppose that Y is δ–hyperbolic. Then there is a δ' depending on δ, λ, and ϵ such that X is δ'–hyperbolic.*

PROOF. Let α_0, α_1, and α_2 be a geodesic triangle in X. For each i, let β_i be a geodesic in Y connecting the endpoints of $\phi(\alpha_i)$. Since $\phi(\alpha_i)$ is a (λ, ϵ)–quasigeodesic in Y, Theorem 4.2.3 provides a μ depending on δ, λ, and ϵ such that $d_{Haus}(\beta_i, \phi(\alpha_i)) \leq \mu$. Since Y is δ–hyperbolic, the geodesic triangle β_0, β_1, β_2 is δ–thin. Combining these two facts, every point $\phi(\alpha_i)$ is within $\mu + 2\delta$ of a point of $\phi(\alpha_{i+1}) \cup \phi(\alpha_{i+2})$. Since these points are in the image of the quasiisometric embedding ϕ, they came from points of α_i and $\alpha_{i+1} \cup \alpha_{i+2}$ whose distance is at most $\delta' := \lambda(\mu + 2\delta + \epsilon)$. □

COROLLARY 4.3.2. *If X and Y are quasiisometric geodesic metric space then X is hyperbolic if and only if Y is hyperbolic.*

Combined with Theorem 3.4.2, this gives:

COROLLARY 4.3.3. *Let G be a finitely generated group. The following are equivalent:*

- *For some finite generating set S of G, $\mathrm{Cay}(G, S)$ is hyperbolic.*
- *For every finite generating set S of G, $\mathrm{Cay}(G, S)$ is hyperbolic.*
- *For some proper geodesic metric space X admitting a geometric G–action, X is hyperbolic.*
- *For every proper geodesic metric space X admitting a geometric G–action, X is hyperbolic.*

In light of Corollary 4.3.3, we make the following definition:

DEFINITION 4.3.4. A finitely generated group G is *hyperbolic* if $\mathrm{Cay}(G, S)$ is hyperbolic for some, equivalently, for every, finite generating set S.

EXAMPLE 4.3.5.

(a) Every finite group is hyperbolic, since its Cayley graph is bounded.
(b) Every finitely generated free group is hyperbolic. With respect to a free generating set, its Cayley graph is a tree, hence is 0–hyperbolic.
(c) The surface group Γ_g for $g \geq 2$ is hyperbolic, since it acts geometrically on the hyperbolic space $\mathbb{H}^2$. More generally, the fundamental group of any closed hyperbolic manifold is hyperbolic.
(d) If H is a hyperbolic group and F is a finite group then $H \times F$ is a hyperbolic group, since it is quasiisometric to H, by Example 3.3.6.

4.4. Exercises

EXERCISE 4.1 (The four-point condition). Let X be a δ–hyperbolic geodesic metric space. For $w, x, y, z \in X$, consider the three numbers:

$$d(w,x) + d(y,z)$$
$$d(w,y) + d(x,z)$$
$$d(w,z) + d(x,y)$$

(a) Use Proposition 4.1.6 to show that the two largest of these numbers differ by at most 8δ.

(b) Conversely, suppose that X is any metric space in which the two largest of these three numbers differ by at most K, for every choice of w, x, y, z. Show that its Gromov products satisfy:

$$(x \mid z)_w \geq \min\{(x \mid y)_w, (y \mid z)_w\} - K/2$$

EXERCISE 4.2 (Gromov products and opposite sides). Let X be a δ–hyperbolic geodesic metric space. Let $o, x, y \in X$, and let $[x, y]$ be a geodesic segment. Set $D := (x \mid y)_o$. Show:

$$d(o, [x, y]) \leq D + 3\delta$$

Let $c \in [o, x]$ be the point with $d(o, c) = D$. Show that c is at distance at most 4δ from each of the three sides of the triangle. Conclude that every geodesic triangle in X has a point at distance at most 4δ from all three sides.

EXERCISE 4.3. For each $R > 0$, let γ_R be the piecewise geodesic in $\mathbb{R}^2$ joining $(-R, 0)$ to $(0, R)$ to $(R, 0)$, parametrized by arclength. Prove that γ_R is a $(\sqrt{2}, 0)$–quasigeodesic and that its Hausdorff distance from the straight segment joining its endpoints is R. Explain why this does not contradict the Morse lemma.

EXERCISE 4.4 (Products are nonhyperbolic). Prove that if $G = A \times B$ is a product of finitely generated, infinite groups, then G is not hyperbolic.

EXERCISE 4.5 (A free product acting on a tree). Consider the finitely generated group:

$$G := \mathbb{Z}/3\mathbb{Z} * \mathbb{Z}/2\mathbb{Z} = \langle a, b \mid a^3, b^2 \rangle$$

Show it acts geometrically on the regular 3–valent tree. Conclude that G is a hyperbolic group.

Hint: Use the reduced-word normal form for free products: every element in G can be expressed uniquely as a word of the form $a^{x_1} b a^{x_2} b \cdots b a^{x_n} b^y$, where $y \in \{0, 1\}$, $x_1 \in \{0, 1, 2\}$, and $x_i \in \{1, 2\}$ for all $1 < i \leq n$.

LECTURE 5

The boundary of a hyperbolic space

LECTURE 6

Algebraic consequences of hyperbolicity

Part 3

Beyond hyperbolicity

LECTURE 7

Relatively hyperbolic groups

LECTURE 8

Acylindrically hyperbolic groups

LECTURE 9

TBD

Bibliography

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